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JEE MAIN · CHEMISTRY · CHAPTER-WISE PYQS

Some Basic Concepts of Chemistry

JEE Main Chemistry · Previous Year Questions with Answer Key

259	179	2002 – 2026	5	English
QUESTIONS	PAPERS COVERED	YEARS	OUT OF SYLLABUS	LANGUAGE

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MCQ (Single Correct Answer) 130 questions

Q1. JEE Main 2026 (Online) 8th April Evening Shift

Open solution

Match List - I with List - II.

List - I Mass of substance		List - II Number of atoms	
A.	1.8 mg water	I.	$2 \times 10^{-4} \times N_A$
B.	9.8 mg sulphuric acid	II.	$1.5 \times 10^{-4} \times N_A$
C.	1.8 mg carbon	III.	$3 \times 10^{-4} \times N_A$
D.	5.85 mg salt $(NaCl)$	IV.	$7 \times 10^{-4} \times N_A$

Choose the correct answer from the options given below :

(A) A-IV, B-III, C-I, D-II **(B)** A-III, B-II, C-IV, D-I

(C) A-III, B-IV, C-II, D-I **(D)** A-III, B-IV, C-I, D-II

Q2. JEE Main 2026 (Online) 6th April Evening Shift

Open solution

Which of the following contain the same number of atoms?

(Given : Molar mass in $gmol^{-1}$ of H, He, O and S are 1, 4, 16 and 32 respectively)

A. 2 g of O_2 gas

B. 4 g of SO_2 gas

C. 1400 mL of O_2 at STP

D. 0.05 L of He at STP

E. 0.0625 mol of H_2 gas

Choose the correct answer from the options given below :

(A) A and B only **(B)** B and C only **(C)** C and D only **(D)** A, C and E only

Q3. JEE Main 2026 (Online) 6th April Morning Shift

Open solution

An oxide of iron contains 69.9% iron, its empirical formula, is :

(Given : Molar mass of Fe and O are 56 and $16\ g\ mol^{-1}$ respectively.)

(A) FeO **(B)** Fe_2O_3 **(C)** Fe_3O_4 **(D)** FeO_3

Q4. JEE Main 2026 (Online) 5th April Evening Shift

Open solution

What volume of hydrogen gas at STP would be liberated by action of 50 mL of H_2SO_4 of 50% purity (density $= 1.3\ g\ mL^{-1}$) on 20 g of zinc?

Given : Molar mass of H, O, S, Zn are 1, 16, 32, $65\ g\ mol^{-1}$ respectively.

(A) 5.824 L **(B)** 7.428 L **(C)** 6.892 L **(D)** 8.375 L

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Q5. JEE Main 2026 (Online) 5th April Morning Shift

How many grams of residue is obtained by heating 2.76 g of silver carbonate?

(Given : Molar mass of C, O and Ag are 12, 16 and 108 g mol^{-1} respectively)

(A) 1.08 g (B) 2.16 g (C) 3.24 g (D) 4.32 g

Open solution

Q6. JEE Main 2026 (Online) 5th April Morning Shift

What is the mole fraction of water in 10% by weight (w/w) of aqueous urea solution?

[Given: Molar mass of H, O, C and N are 1, 16, 12 and 14 g mol^{-1} respectively.]

(A) 0.825 (B) 0.032 (C) 0.867 (D) 0.967

Open solution

Q7. JEE Main 2026 (Online) 4th April Evening Shift

The correct order of total number of atoms present in

(A) 2 moles of cyclohexane

(B) 684 g of sucrose

(C) 90.8 L of dihydrogen at STP

is :

(A) C > A > B (B) C > B > A (C) B > C > A (D) B > A > C

Open solution

Q8. JEE Main 2026 (Online) 4th April Morning Shift

Number of moles and number of molecules in 1.4187 L of SO_2 at STP respectively are

(A) $0.1266; 3.812 \times 10^{22}$ (B) $0.0633; 3.812 \times 10^{22}$

(C) $0.1266; 7.6238 \times 10^{22}$ (D) $0.0633; 7.6238 \times 10^{22}$

Open solution

Q9. JEE Main 2026 (Online) 2nd April Evening Shift

The ratio of mass percentage (w/w) of C : H in a hydrocarbon is 12 : 1. It has two carbon atoms. The weight (in g) of CO_2(g) formed when 3.38 g of this hydrocarbon is completely burnt in oxygen is :

(Given : Molar mass in g mol^{-1} C : 12, H : 1, O : 16)

(A) 5.68 (B) 11.44 (C) 22.74 (D) 17.05

Open solution

Q10. JEE Main 2026 (Online) 2nd April Morning Shift

The mass of iron converted into Fe_3O_4 by the action of 18 g of steam is:

(Given: Molar mass of H, O and Fe are 1, 16 and 56 g mol^{-1} respectively)

Assume iron is present in excess:

(A) 2.1 g (B) 4.2 g (C) 21 g (D) 42 g

Open solution

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Q11. JEE Main 2026 (Online) 28th January Evening Shift

Open solution

For the given reaction;

$CaCO_3 + 2HCl \rightarrow CaCl_2 + H_2O + CO_2$

If 90 g $CaCO_3$ is added to 300 mL of HCl which contains 38.55% HCl by mass and has density 1.13 g mL^{-1}, then which of the following option is **correct**?

Given molar mass of H, Cl, Ca and O are 1, 35.5, 40 and 16 g mol^{-1} respectively.

(A) 64.97 g of HCl remains unreacted

(B) 60.32 g of HCl remains unreacted

(C) 32.85 g of $CaCO_3$ remains unreacted

(D) 97.30 g of HCl reacted

Q12. JEE Main 2026 (Online) 22nd January Evening Shift

Open solution

At $\mathrm{T(K)}$, 100 g of $98\%\mathrm{H_2SO_4(w/w)}$ aqueous solution is mixed with 100 g of $49\%\mathrm{H_2SO_4(w/w)}$ aqueous solution. What is the mole fraction of $\mathrm{H_2SO_4}$ in the resultant solution?

(Given : Atomic mass $\mathrm{H = 1u; S = 32u; O = 16u}$).

(Assume that temperature after mixing remains constant)

(A) 0.9

(B) 0.663

(C) 0.1

(D) 0.337

Q13. JEE Main 2026 (Online) 22nd January Evening Shift

Open solution

$\mathrm{A + 2\,B \longrightarrow AB_2}$

36.0 g of 'A' (Molar mass : $60\ \mathrm{g\ mol^{-1}}$) and 56.0 g of ' B ' (Molar mass : $80\ \mathrm{g\ mol^{-1}}$) are allowed to react. Which of the following statements are correct ?

A. 'A' is the limiting reagent.

B. 77.0 g of $\mathrm{AB_2}$ is formed.

C. Molar mass of $\mathrm{AB_2}$ is $140\ \mathrm{g\ mol^{-1}}$.

D. 15.0 g of A is left unreacted after the completion of reaction.

Choose the correct answer from the options given below :

(A) A and C Only

(B) A and B Only

(C) C and D Only

(D) B and D Only

Q14. JEE Main 2026 (Online) 22nd January Morning Shift

Open solution

In the reaction,

$\mathrm{2Al(\ s) + 6HCl(aq) \rightarrow 2Al^{3+}(aq) + 6Cl^{-}(aq) + 3H_2(\ g)}$

(A) $\mathrm{12\ L\ HCl(aq)}$ is consumed for every $\mathrm{6\ L\ H_2(\ g)}$ produced.

(B) $\mathrm{11.2\ L\ H_2(\ g)}$ at STP is produced for every mole of HCl consumed.

(C) $\mathrm{33.6\ L\ H_2(\ g)}$ is produced regardless of temperature and pressure for every mole of Al that reacts.

(D) $\mathrm{67.2\ L\ H_2(\ g)}$ at STP is produced for every mole of Al that reacts.

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Q15. JEE Main 2026 (Online) 21st January Evening Shift

Aqueous HCl reacts with $MnO_2(s)$ to form $MnCl_2(aq)$, $Cl_2(g)$, and $H_2O(l)$. What is the weight (in g) of Cl_2 liberated when 8.7 g of $MnO_2(s)$ is reacted with excess aqueous HCl solution? (Given Molar mass in g mol^{-1} Mn = 55, Cl = 35.5, O = 16, H = 1)

Open solution

(A) 14.2 (B) 21.3 (C) 7.1 (D) 71

Q16. JEE Main 2026 (Online) 21st January Morning Shift

14.0 g of calcium metal is allowed to react with excess HCl at 1.0 atm pressure and 273 K . Which of the following statements is incorrect?

[Given : Molar mass in $gmol^{-1}$ of $Ca-40, Cl-35.5, H-1$]

Open solution

(A) The limiting reagent is calcium metal.
(B) 33.3 g of $CaCl_2$ is produced.
(C) 7.84 L of H_2 gas is evolved.
(D) 0.35 mol of H_2 gas is evolved.

Q17. JEE Main 2026 (Online) 21st January Morning Shift

80 mL of a hydrocarbon on mixing with 264 mL of oxygen in a closed U-tube undergoes complete combustion. The residual gases after cooling to 273 K occupy 224 mL . When the system is treated with KOH solution, the volume decreases to 64 mL . The formula of the hydrocarbon is :

Open solution

(A) C_2H_2 (B) C_2H_4 (C) C_4H_{10} (D) C_2H_6

Q18. JEE Main 2025 (Online) 8th April Evening Shift

On combustion 0.210 g of an organic compound containing C, H and O gave 0.127 g H_2O and 0.307 g CO_2. The percentages of hydrogen and oxygen in the given organic compound respectively are:

Open solution

(A) 7.55, 43.85 (B) 6.72, 53.41 (C) 6.72, 39.87 (D) 53.41, 39.6

Q19. JEE Main 2025 (Online) 3rd April Evening Shift

Mass of magnesium required to produce 220 mL of hydrogen gas at STP on reaction with excess of dil. HCl is

Given: Molar mass of Mg is $24\ g\ mol^{-1}$.

Open solution

(A) 0.24 mg (B) 235.7 g (C) 2.444 g (D) 236 mg

Q20. JEE Main 2025 (Online) 3rd April Evening Shift

10 mL of 2 M NaOH solution is added to 20 mL of 1 M HCl solution kept in a beaker. Now, 10 mL of this mixture is poured into a volumetric flask of 100 mL containing 2 moles of HCl and made the volume upto the mark with distilled water. The solution in this flask is :

Open solution

(A) 0.2 M NaCl solution
(B) 20 M HCl solution
(C) Neutral solution
(D) 10 M HCl solution

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Q21. JEE Main 2025 (Online) 3rd April Morning Shift

Among 10^{-9} g (each) of the following elements, which one will have the highest number of atoms?

Element: Pb, Po, Pr and Pt

(A) Pr (B) Pt (C) Po (D) Pb

Open solution

Q22. JEE Main 2025 (Online) 2nd April Morning Shift

$CaCO_3(s) + 2HCl(aq) \rightarrow CaCl_2(aq) + CO_2(g) + H_2O(l)$

Consider the above reaction, what mass of $CaCl_2$ will be formed if 250 mL of 0.76 M HCl reacts with 1000 g of $CaCO_3$?

(Given : Molar mass of Ca, C, O, H and Cl are $40, 12, 16, 1$ and 35.5 g mol^{-1}, respectively)

(A) 3.908 g (B) 2.636 g (C) 10.545 g (D) 5.272 g

Open solution

Q23. JEE Main 2025 (Online) 2nd April Morning Shift

On complete combustion 1.0 g of an organic compound (X) gave 1.46 g of CO_2 and 0.567 g of H_2O. The empirical formula mass of compound (X) is _____ g. (Given molar mass in $gmol^{-1}$ C : 12, H : 1, O : 16)

(A) 60 (B) 45 (C) 30 (D) 15

Open solution

Q24. JEE Main 2025 (Online) 29th January Morning Shift

Choose the **correct** statements.

(A) Weight of a substance is the amount of matter present in it.

(B) Mass is the force exerted by gravity on an object.

(C) Volume is the amount of space occupied by a substance.

(D) Temperatures below 0°C are possible in Celsius scale, but in Kelvin scale negative temperature is not possible.

(E) Precision refers to the closeness of various measurements for the same quantity.

Choose the **correct** answer from the options given below :

(A) (B), (C) and (D) Only (B) (A), (B) and (C) Only

(C) (C), (D) and (E) Only (D) (A), (D) and (E) Only

Open solution

Q25. JEE Main 2025 (Online) 28th January Evening Shift

Concentrated nitric acid is labelled as 75% by mass. The volume in mL of the solution which contains 30 g of nitric acid is _____.

Given : Density of nitric acid solution is 1.25 g/mL.

(A) 32 (B) 40 (C) 55 (D) 45

Open solution

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Q26. JEE Main 2025 (Online) 24th January Evening Shift

The elemental composition of a compound is $54.2\%\mathrm{C}, 9.2\%\mathrm{H}$ and $36.6\%\mathrm{O}$. If the molar mass of the compound is $132\ \mathrm{g\ mol^{-1}}$, the molecular formula of the compound is : [Given : The relative atomic mass of $\mathrm{C : H : O} = 12 : 1 : 16$]

(A) $\mathrm{C_4H_8O_2}$ (B) $\mathrm{C_6H_{12}O_6}$

(C) $\mathrm{C_4H_9O_3}$ (D) $\mathrm{C_6H_{12}O_3}$

Open solution

Q27. JEE Main 2025 (Online) 23rd January Morning Shift

$2.8 \times 10^{-3}\ \mathrm{mol}$ of $\mathrm{CO_2}$ is left after removing 10^{21} molecules from its ' x ' mg sample. The mass of $\mathrm{CO_2}$ taken initially is Given: $\mathrm{N_A} = 6.02 \times 10^{23}\ \mathrm{mol^{-1}}$

(A) 98.3 mg (B) 196.2 mg (C) 150.4 mg (D) 48.2 mg

Open solution

Q28. JEE Main 2025 (Online) 22nd January Evening Shift

Density of 3 M NaCl solution is $1.25\ \mathrm{g/mL}$. The molality of the solution is :

(A) 2.79 m (B) 2 m (C) 1.79 m (D) 3 m

Open solution

Q29. JEE Main 2024 (Online) 9th April Evening Shift

The candela is the luminous intensity, in a given direction, of a source that emits monochromatic radiation of frequency 'A' $\times 10^{12}$ hertz and that has a radiant intensity in that direction of $\frac{1}{'B'}$, watt per steradian. 'A' and 'B' are respectively

(A) 450 and 683 (B) 540 and $\frac{1}{683}$ (C) 540 and 683 (D) 450 and $\frac{1}{683}$

Open solution

Q30. JEE Main 2024 (Online) 8th April Morning Shift

Combustion of glucose $(\mathrm{C_6H_{12}O_6})$ produces $\mathrm{CO_2}$ and water. The amount of oxygen (in g) required for the complete combustion of $900\ \mathrm{g}$ of glucose is :

[Molar mass of glucose in $\mathrm{g\ mol^{-1}} = 180$]

(A) 32 (B) 480 (C) 800 (D) 960

Open solution

Q31. JEE Main 2024 (Online) 6th April Evening Shift

Molality (m) of $3\mathrm{M}$ aqueous solution of NaCl is : (Given : Density of solution $= 1.25\ \mathrm{g\ mL^{-1}}$, Molar mass in $\mathrm{g\ mol^{-1}}$: $\mathrm{Na} - 23, \mathrm{Cl} - 35.5$)

(A) 2.90 m (B) 3.85 m (C) 1.90 m (D) 2.79 m

Open solution

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Q32. JEE Main 2024 (Online) 6th April Morning Shift

The density of 'x' M solution ('x' molar) of NaOH is $1.12\ \mathrm{g\ mL^{-1}}$, while in molality, the concentration of the solution is $3\ \mathrm{m}$ (3 molal). Then x is

(Given : Molar mass of NaOH is $40\ \mathrm{g/mol}$)

Open solution

(A) 3.8 (B) 3.0 (C) 3.5 (D) 2.8

Q33. JEE Main 2024 (Online) 5th April Evening Shift

The number of moles of methane required to produce $11\ \mathrm{g}\ \mathrm{CO_2(g)}$ after complete combustion is : (Given molar mass of methane in $\mathrm{g\ mol^{-1}}$: 16)

Open solution

(A) 0.35 (B) 0.25 (C) 0.5 (D) 0.75

Q34. JEE Main 2024 (Online) 5th April Morning Shift

The incorrect postulates of the Dalton's atomic theory are :

(A) Atoms of different elements differ in mass.

(B) Matter consists of divisible atoms.

(C) Compounds are formed when atoms of different elements combine in a fixed ratio.

(D) All the atoms of given element have different properties including mass.

(E) Chemical reactions involve reorganisation of atoms.

Choose the correct answer from the options given below :

Open solution

(A) (B), (D) only (B) (A), (B), (D) only

(C) (B), (D), (E) only (D) (C), (D), (E) only

Q35. JEE Main 2024 (Online) 5th April Morning Shift

An organic compound has 42.1% carbon, 6.4% hydrogen and the remainder is oxygen. If its molecular weight is 342 , then its molecular formula is :

Open solution

(A) $\mathrm{C_{14}H_{20}O_{10}}$ (B) $\mathrm{C_{12}H_{22}O_{11}}$

(C) $\mathrm{C_{12}H_{20}O_{12}}$ (D) $\mathrm{C_{11}H_{18}O_{12}}$

Q36. JEE Main 2024 (Online) 4th April Morning Shift

The Molarity (M) of an aqueous solution containing $5.85\ \mathrm{g}$ of NaCl in $500\ \mathrm{mL}$ water is : (Given : Molar Mass Na : 23 and Cl : $35.5\ \mathrm{gmol^{-1}}$)

Open solution

(A) 20 (B) 4 (C) 2 (D) 0.2

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Q37. JEE Main 2024 (Online) 31st January Evening Shift

A sample of $CaCO_3$ and $MgCO_3$ weighed $2.21\ g$ is ignited to constant weight of $1.152\ g$. The composition of mixture is :

(Given molar mass in $g\ mol^{-1} CaCO_3 : 100, MgCO_3 : 84$)

(A) $1.187\ g\ CaCO_3 + 1.187\ g\ MgCO_3$

(B) $1.187\ g\ CaCO_3 + 1.023\ g\ MgCO_3$

(C) $1.023\ g\ CaCO_3 + 1.187\ g\ MgCO_3$

(D) $1.023\ g\ CaCO_3 + 1.023\ g\ MgCO_3$

Open solution

Q38. JEE Main 2024 (Online) 30th January Evening Shift

If a substance 'A' dissolves in solution of a mixture of 'B' and 'C' with their respective number of moles as n_A, n_B and n_{C_3}. Mole fraction of C in the solution is

(A) $\frac{n_C}{n_A \times n_B \times n_C}$

(B) $\frac{n_B}{n_A + n_B}$

(C) $\frac{n_C}{n_A + n_B + n_C}$

(D) $\frac{n_C}{n_A - n_B - n_C}$

Open solution

Q39. JEE Main 2024 (Online) 27th January Evening Shift

The quantity which changes with temperature is :

(A) Molality (B) Molarity (C) Mole fraction (D) Mass percentage

Open solution

Q40. JEE Main 2023 (Online) 12th April Morning Shift

A metal chloride contains 55.0% of chlorine by weight . $100\ mL$ vapours of the metal chloride at STP weigh $0.57\ g$. The molecular formula of the metal chloride is

(Given: Atomic mass of chlorine is $35.5u$)

(A) MCl_2 (B) MCl_4 (C) MCl_3 (D) MCl

Open solution

Q41. JEE Main 2023 (Online) 11th April Evening Shift

A solution is prepared by adding $2\ g$ of "X" to 1 mole of water. Mass percent of "X" in the solution is :

(A) 20% (B) 10% (C) 2% (D) 5%

Open solution

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Q42. JEE Main 2023 (Online) 10th April Evening Shift

Open solution

Given below are two statements: one is labelled as Assertion $\mathbf{A}$ and the other is labelled as Reason $\mathbf{R}$

Assertion A : $3.1500\ \mathrm{g}$ of hydrated oxalic acid dissolved in water to make $250.0\ \mathrm{mL}$ solution will result in $0.1\ \mathrm{M}$ oxalic acid solution.

Reason $\mathbf{R}$: Molar mass of hydrated oxalic acid is $126\ \mathrm{g\ mol^{-1}}$

In the light of the above statements, choose the correct answer from the options given below.

(A) Both A and R are true but R is NOT the correct explanation of A

(B) A is true but R is false

(C) Both A and R are true and R is the correct explanation of A

(D) A is false but R is true

Q43. JEE Main 2023 (Online) 10th April Evening Shift

Open solution

Match List I with List II

	List - I		List - II
A.	16 g of $\mathrm{CH_4\ (g)}$	I.	Weighs 28 g
B.	1 g of $\mathrm{H_2\ (g)}$	II.	60.2×10^{23} electrons
C.	1 mole of $\mathrm{N_2\ (g)}$	III.	Weighs 32 g
D.	0.5 mol of $\mathrm{SO_2\ (g)}$	IV.	Occupies 11.4 L volume at STP

Choose the correct answer from the options given below:

(A) A-II, B-III, C-IV, D-I

(B) A-II, B-IV, C-I, D-III

(C) A-I, B-III, C-II, D-IV

(D) A-II, B-IV, C-III, D-I

Q44. JEE Main 2023 (Online) 10th April Morning Shift

Open solution

The number of molecules and moles in 2.8375 litres of O_2 at STP are respectively

(A) 1.505×10^{23} and 0.250 mol

(B) 7.527×10^{22} and 0.250 mol

(C) 7.527×10^{23} and 0.125 mol

(D) 7.527×10^{22} and 0.125 mol

Q45. JEE Main 2023 (Online) 6th April Evening Shift

Open solution

The volume of $0.02\ \mathrm{M}$ aqueous HBr required to neutralize $10.0\ \mathrm{mL}$ of $0.01\ \mathrm{M}$ aqueous $\mathrm{Ba(OH)_2}$ is (Assume complete neutralization)

(A) 7.5 mL

(B) 5.0 mL

(C) 10.0 mL

(D) 2.5 mL

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Q46. JEE Main 2023 (Online) 31st January Evening Shift

When a hydrocarbon A undergoes complete combustion it requires 11 equivalents of oxygen and produces 4 equivalents of water. What is the molecular formula of A ?

(A) C_9H_8 (B) C_5H_8 (C) $C_{11}H_4$ (D) $C_{11}H_8$

Open solution

Q47. JEE Main 2023 (Online) 29th January Evening Shift

When a hydrocarbon A undergoes combustion in the presence of air, it requires 9.5 equivalents of oxygen and produces 3 equivalents of water. What is the molecular formula of A?

(A) C_9H_6 (B) C_6H_6 (C) C_8H_6 (D) C_9H_9

Open solution

Q48. JEE Main 2023 (Online) 25th January Evening Shift

What is the mass ratio of ethylene glycol ($C_2H_6O_2$, molar mass = 62 g/mol) required for making 500 g of 0.25 molal aqueous solution and 250 mL of 0.25 molar aqueous solution?

(A) 1 : 2 (B) 1 : 1 (C) 2 : 1 (D) 3 : 1

Open solution

Q49. JEE Main 2023 (Online) 25th January Morning Shift

'25 volume' hydrogen peroxide means

(A) 1 L marketed solution contains 75 g of H_2O_2.

(B) 1 L marketed solution contains 250 g of H_2O_2.

(C) 1 L marketed solution contains 25 g of H_2O_2.

(D) 100 mL marketed solution contains 25 g of H_2O_2.

Open solution

Q50. JEE Main 2022 (Online) 29th July Evening Shift

Consider the reaction

$4HNO_3(l) + 3KCl(s) \rightarrow Cl_2(g) + NOCl(g) + 2H_2O(g) + 3KNO_3(s)$

The amount of HNO_3 required to produce 110.0 g of KNO_3 is

(Given: Atomic masses of H, O, N and K are $1, 16, 14$ and 39, respectively.)

(A) 32.2 g (B) 69.4 g (C) 91.5 g (D) 162.5 g

Open solution

Q51. JEE Main 2022 (Online) 29th July Evening Shift

$C(s) + O_2(g) \rightarrow CO_2(g) + 400\ kJ$

$C(s) + \frac{1}{2}O_2(g) \rightarrow CO(g) + 100\ kJ$

When coal of purity 60% is allowed to burn in presence of insufficient oxygen, 60% of carbon is converted into 'CO' and the remaining is converted into 'CO_2'. The heat generated when 0.6 kg of coal is burnt is _____.

(A) 1600 kJ (B) 3200 kJ (C) 4400 kJ (D) 6600 kJ

Open solution

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Q52. JEE Main 2022 (Online) 29th July Morning Shift

$N_{2(g)} + 3H_{2(g)} \rightleftharpoons 2NH_{3(g)}$

$20\ g \quad 5\ g$

Consider the above reaction, the limiting reagent of the reaction and number of moles of NH_3 formed respectively are :

(A) $H_2, 1.42$ moles (B) $H_2, 0.71$ moles (C) $N_2, 1.42$ moles (D) $N_2, 0.71$ moles

Open solution

Q53. JEE Main 2022 (Online) 27th July Morning Shift

$250\ g$ solution of D-glucose in water contains 10.8% of carbon by weight. The molality of the solution is nearest to

(Given: Atomic Weights are, $H, 1\ u; C, 12\ u; O, 16\ u$)

(A) 1.03 (B) 2.06 (C) 3.09 (D) 5.40

Open solution

Q54. JEE Main 2022 (Online) 26th July Evening Shift

Hemoglobin contains 0.34% of iron by mass. The number of Fe atoms in $3.3\ g$ of hemoglobin is

(Given: Atomic mass of Fe is $56\ u$, $N_A = 6.022 \times 10^{23}\ mol^{-1}$.)

(A) 1.21×10^5 (B) 12.0×10^{16} (C) 1.21×10^{20} (D) 3.4×10^{22}

Open solution

Q55. JEE Main 2022 (Online) 25th July Morning Shift

SO_2Cl_2 on reaction with excess of water results into acidic mixture

$SO_2Cl_2 + 2H_2O \rightarrow H_2SO_4 + 2HCl$

16 moles of $NaOH$ is required for the complete neutralisation of the resultant acidic mixture. The number of moles of SO_2Cl_2 used is :

(A) 16 (B) 8 (C) 4 (D) 2

Open solution

Q56. JEE Main 2022 (Online) 29th June Morning Shift

Production of iron in blast furnace follows the following equation

$Fe_3O_4(s) + 4CO(g) \rightarrow 3Fe(l) + 4CO_2(g)$

when 4.640 kg of Fe_3O_4 and 2.520 kg of CO are allowed to react then the amount of iron (in g) produced is :

[Given : Molar Atomic mass (g mol^{-1}) : Fe = 56, Molar Atomic mass (g mol^{-1}) : O = 16, Molar Atomic mass (g mol^{-1}) : C = 12]

(A) 1400 (B) 2200 (C) 3360 (D) 4200

Open solution

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Q57. JEE Main 2022 (Online) 28th June Evening Shift

Compound A contains 8.7% Hydrogen, 74% Carbon and 17.3% Nitrogen. The molecular formula of the compound is,

Given : Atomic masses of C, H and N are 12, 1 and 14 amu respectively.

The molar mass of the compound A is 162 g mol^{-1}.

(A) $C_4H_6N_2$ (B) C_2H_3N (C) C_5H_7N (D) $C_{10}H_{14}N_2$

Open solution

Q58. JEE Main 2022 (Online) 26th June Morning Shift

A commercially sold conc. HCl is 35% HCl by mass. If the density of this commercial acid is 1.46 g/mL, the molarity of this solution is:

(Atomic mass : Cl = 35.5 amu, H = 1 amu)

(A) 10.2 M (B) 12.5 M (C) 14.0 M (D) 18.2 M

Open solution

Q59. JEE Main 2022 (Online) 24th June Evening Shift

120 g of an organic compound that contains only carbon and hydrogen gives 330 g of CO_2 and 270 g of water on complete combustion. The percentage of carbon and hydrogen, respectively are

(A) 25 and 75 (B) 40 and 60 (C) 60 and 40 (D) 75 and 25

Open solution

Q60. JEE Main 2022 (Online) 24th June Morning Shift

If a rocket runs on a fuel ($C_{15}H_{30}$) and liquid oxygen, the weight of oxygen required and CO_2 released for every litre of fuel respectively are :

(Given : density of the fuel is 0.756 g/mL)

(A) 1188 g and 1296 g (B) 2376 g and 2592 g
(C) 2592 g and 2376 g (D) 3429 g and 3142 g

Open solution

Q61. JEE Main 2021 (Online) 16th March Evening Shift

The exact volumes of 1 M NaOH solution required to neutralise 50 mL of 1 M H_3PO_3 solution and 100 mL of 2 M H_3PO_2 solution, respectively, are :

(A) 100 mL and 50 mL (B) 100 mL and 200 mL
(C) 100 mL and 100 mL (D) 50 mL and 50 mL

Open solution

Q62. JEE Main 2021 (Online) 25th February Morning Shift

Complete combustion of 1.80 g of an oxygen containing compound ($C_xH_yO_z$) gave 2.64 g of CO_2 and 1.08 g of H_2O. The percentage of oxygen in the organic compound is :

(A) 51.63 (B) 50.33 (C) 63.53 (D) 53.33

Open solution

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Q63. JEE Main 2020 (Online) 6th September Evening Slot

The average molar mass of chlorine is 35.5 g mol^{-1}. The ratio of ^{35}Cl to ^{37}Cl in naturally occuring chlorine is close to :

(A) 1 : 1 (B) 2 : 1 (C) 3 : 1 (D) 4 : 1

Open solution

Q64. JEE Main 2020 (Online) 6th September Morning Slot

A solution of two components containing n_1 moles of the 1st component and n_2 moles of the 2nd component is prepared. M_1 and M_2 are the molecular weights of component 1 and 2 respectively. If d is the density of the solution in g mL^{-1}, C_2 is the molarity and x_2 is the mole fraction of the 2nd component, then C_2 can be expressed as :

(A) $C_2 = \frac{1000x_2}{M_1+x_2(M_2-M_1)}$ (B) $C_2 = \frac{1000dx_2}{M_1+x_2(M_2-M_1)}$

(C) $C_2 = \frac{dx_2}{M_1+x_2(M_2-M_1)}$ (D) $C_2 = \frac{dx_1}{M_1+x_2(M_2-M_1)}$

Open solution

Q65. JEE Main 2020 (Online) 3rd September Evening Slot

The strengths of 5.6 volume hydrogen peroxide (of density 1 g/mL) in terms of mass percentage and molarity (M), respectively, are: (Take molar mass of hydrogen peroxide as 34 g/mol)

(A) 0.85 and 0.5 (B) 0.85 and 0.25 (C) 1.7 and 0.25 (D) 1.7 and 0.5

Open solution

Q66. JEE Main 2020 (Online) 7th January Evening Slot

The ammonia (NH_3) released on quantitative reaction of 0.6 g urea (NH_2CONH_2) with sodium hydroxide (NaOH) can be neutralized by :

(A) 200 ml of 0.02 N HCl (B) 100 ml of 0.2 N HCl

(C) 100 ml of 0.1 N HCl (D) 200 ml of 0.4 N HCl

Open solution

Q67. JEE Main 2020 (Online) 7th January Morning Slot

Amongst the following statements, that which was not proposed by Dalton was :

(A) Matter consists of indivisible atoms.

(B) Chemical reactions involve reorganization of atoms. These are neither created nor destroyed in a chemical reaction.

(C) When gases combine or are produced in a chemical reaction they do so in a simple ratio by volume provided all gases are at the same T & P.

(D) All the atoms of a given element have identical properties including identical mass. Atoms of different elements differ in mass.

Open solution

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Q68. JEE Main 2019 (Online) 12th April Evening Slot

25 g of an unknown hydrocarbon upon burning produces 88 g of CO_2 and 9 g of H_2O. This unknown hydrocarbon contains :

(A) 18 g of carbon and 7 g of hydrogen
(B) 20 g of carbon and 5 g of hydrogen
(C) 22 g of carbon and 3 g of hydrogen
(D) 24 g of carbon and 1 g of hydrogen

Open solution

Q69. JEE Main 2019 (Online) 12th April Morning Slot

The mole fraction of a solvent in aqueous solution of a solute is 0.8. The molality (in mol kg^{-1}) of the aqueous solution is :

(A) 13.88×10^{-1}
(B) 13.88×10^{-3}
(C) 13.88
(D) 13.88×10^{-2}

Open solution

Q70. JEE Main 2019 (Online) 12th April Morning Slot

5 moles of AB_2 weigh 125×10^{-3} kg and 10 moles of A_2B_2 weigh 300×10^{-3} kg. The molar mass of A (M_A) and molar mass of B (M_B) in kg mol^{-1} are:

(A) $M_A = 10 \times 10^{-3}$ and $M_B = 5 \times 10^{-3}$
(B) $M_A = 25 \times 10^{-3}$ and $M_B = 50 \times 10^{-3}$
(C) $M_A = 5 \times 10^{-3}$ and $M_B = 10 \times 10^{-3}$
(D) $M_A = 50 \times 10^{-3}$ and $M_B = 25 \times 10^{-3}$

Open solution

Q71. JEE Main 2019 (Online) 10th April Evening Slot

The minimum amount of $O_2(g)$ consumed per gram of reactant is for the reaction :
(Given atomic mass : Fe = 56, O = 16, Mg = 24, P = 31, C = 12, H = 1)

(A) $4Fe(s) + 3O_2(g) \rightarrow 2Fe_2O_3(s)$
(B) $P_4(s) + 5O_2(g) \rightarrow P_4O_{10}(s)$
(C) $C_3H_8(g) + 5O_2(g) \rightarrow 3CO_2(g) + 4H_2O(l)$
(D) $2Mg(s) + O_2(g) \rightarrow 2MgO(s)$

Open solution

Q72. JEE Main 2019 (Online) 10th April Morning Slot

At 300 K and 1 atmospheric pressure, 10 mL of a hydrocarbon required 55 mL of O_2 for complete combustion, and 40 mL of CO_2 is formed. The formula of the hydrocarbon is :

(A) C_4H_7Cl
(B) C_4H_6
(C) C_4H_8
(D) C_4H_{10}

Open solution

Q73. JEE Main 2019 (Online) 9th April Evening Slot

What would be the molality of 20% (mass/ mass) aqueous solution of KI?
(molar mass of KI = 166 g mol^{-1})

(A) 1.51
(B) 1.35
(C) 1.08
(D) 1.48

Open solution

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Q74. JEE Main 2019 (Online) 9th April Morning Slot

For a reaction,
$N_2(g) + 3H_2(g) \rightarrow 2NH_3(g)$;
identify dihydrogen (H_2) as a limiting reagent in the following reaction mixtures.

(A) 56g of N_2 + 10g of H_2
(B) 14g of N_2 + 4g of H_2
(C) 28g of N_2 + 6g of H_2
(D) 35g of N_2 + 8g of H_2

Open solution

Q75. JEE Main 2019 (Online) 8th April Evening Slot

The percentage composition of carbon by mole in methane is :

(A) 80%
(B) 20%
(C) 75%
(D) 25%

Open solution

Q76. JEE Main 2019 (Online) 12th January Evening Slot

8 g of NaOH is dissolved in 18g of H_2O. Mole fraction of NaOH in solution and molality (in mol kg^{-1}) of the solution respectively are -

(A) 0.2, 11.11
(B) 0.167, 22.20
(C) 0.167, 11.11
(D) 0.2, 22.20

Open solution

Q77. JEE Main 2019 (Online) 11th January Evening Slot

25 ml of the given HCl solution requires 30 mL of 0.1 M sodium carbonate solution. What is the volume of this HCl solution required to titrate 30 mL of 0.2 M aqueous NaOH solutions ?

(A) 50 mL
(B) 12.5 mL
(C) 25 mL
(D) 75 mL

Open solution

Q78. JEE Main 2019 (Online) 11th January Morning Slot

A 10 mg effervescent tablet containing sodium bicarbonate and oxalic acid releases 0.25 ml of CO_2 at T = 298.15 K and p = 1 bar. If molar volume of CO_2 is 25.0 L under such condition, what is the percentage of sodium bicarbonate in each tablet ?
[Molar mass of $NaHCO_3$ = 84 g mol^{-1}]

(A) 33.6
(B) 0.84
(C) 8.4
(D) 16.8

Open solution

Q79. JEE Main 2019 (Online) 11th January Morning Slot

An organic compound is estimated through Dumas method and was found to evolve 6 moles of CO_2, 4 moles of H_2O and 1 mole of nitrogen gas. The formula of the compound is :

(A) C_6H_8N
(B) $C_6H_8N_2$
(C) $C_{12}H_8N$
(D) $C_{12}H_8N_2$

Open solution

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Q80. JEE Main 2019 (Online) 10th January Evening Slot

The amount of sugar ($C_{12}H_{22}O_{11}$) required to prepare 2L of its 0.1 M aqueous solution is :

(A) 17.1 g (B) 34.2 g (C) 68.4 g (D) 136.8 g

Open solution

Q81. JEE Main 2019 (Online) 9th January Evening Slot

For the following reaction, in the mass of water produced from 445 g of $C_{57}H_{110}O_6$ is :

$2C_{57}H_{110}O_6(s) + 163\ O_2(g) \rightarrow 114\ CO_2(g) + 110\ H_2O(l)$

(A) 490 g (B) 445 g (C) 495 g (D) 890 g

Open solution

Q82. JEE Main 2019 (Online) 9th January Morning Slot

A solution of sodium sulphate contains 92 g of Na^+ ions per kilogram of water. The molality of Na^+ ions in that solution in mol kg^{-1} is :

(A) 12 (B) 4 (C) 8 (D) 16

Open solution

Q83. JEE Main 2018 (Online) 16th April Morning Slot

An unknown chlorohydrocarbon has 3.55% of chlorine. If each molecule of the hydrocarbon has one chlorine atom only; chlorine atoms present in 1 g of chlorohydrocarbon are :
(Atomic wt. of Cl = 35.5 u; Avogadro constant = 6.023×10^{23} mol^{-1})

(A) 6.023×10^{20} (B) 6.023×10^{9} (C) 6.023×10^{21} (D) 6.023×10^{23}

Open solution

Q84. JEE Main 2018 (Offline)

The ratio of mass percent of C and H of an organic compound ($C_XH_YO_Z$) is 6 : 1. If one molecule of the above compound ($C_XH_YO_Z$) contains half as much oxygen as required to burn one molecule of compound C_XH_Y completely to CO_2 and H_2O. The empirical formula of compound $C_XH_YO_Z$ is

(A) $C_2H_4O_3$ (B) $C_3H_6O_3$ (C) C_2H_4O (D) $C_3H_4O_2$

Open solution

Q85. JEE Main 2018 (Online) 15th April Evening Slot

For per gram of reactant, the maximum quantity of N_2 gas is produced in which of the following thermal decomposition reactions ?
(Given : Atomic wt. - Cr = 52 u, Ba = 137 u)

(A) $(NH_4)_2Cr_2O_7(s) \rightarrow N_2(g) + 4H_2O(g) + Cr_2O_3(s)$

(B) $2NH_4NO_3(s) \rightarrow 2\ N_2(g) + 4H_2O(g) + O_2(g)$

(C) $Ba(N_3)_2(s) \rightarrow Ba(s) + 3N_2(g)$

(D) $2NH_3(g) \rightarrow N_2(g) + 3H_2(g)$

Open solution

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Q86. JEE Main 2018 (Online) 15th April Morning Slot

A sample of $NaClO_3$ is converted by heat to $NaCl$ with a loss of $0.16\ g$ of oxygen. The residue is dissolved in water and precipitated as $AgCl$. The mass of $AgCl$ (in g) obtained will be : (Given : Molar mass of $AgCl = 143.5\ g\ mol^{-1}$)

(A) 0.35 (B) 0.41 (C) 0.48 (D) 0.54

Open solution

Q87. JEE Main 2017 (Online) 9th April Morning Slot

What quantity (in mL) of a 45% acid solution of a mono-protic strong acid must be mixed with a 20% solution of the same acid to produce 800 mL of a 29.875% acid solution ?

(A) 320 (B) 325 (C) 316 (D) 330

Open solution

Q88. JEE Main 2017 (Online) 8th April Morning Slot

Excess of NaOH (aq) was added to 100 mL of $FeCl_3$ (aq) resulting into 2.14 g of $Fe(OH)_3$. The molarity of $FeCl_3$ (aq) is :

(Given molar mass of Fe = 56 g mol^{-1} and molar mass of Cl = 35.5 g mol^{-1})

(A) 0.2 M (B) 0.3 M (C) 0.6 M (D) 1.8 M

Open solution

Q89. JEE Main 2017 (Offline)

The most abundant elements by mass in the body of a healthy human adult are: Oxygen (61.4%); Carbon (22.9%), Hydrogen (10.0%); and Nitrogen (2.6%). The weight which a 75 kg person would gain if all ^1H atoms are replaced by ^2H atoms is:

(A) 37.5 kg (B) 7.5 kg (C) 10 kg (D) 15 kg

Open solution

Q90. JEE Main 2017 (Offline)

1 gram of a carbonate (M_2CO_3) on treatment with excess HCl produces 0.01186 mole of CO_2. The molar mass of M_2CO_3 in g mol^{-1} is:

(A) 84.3 (B) 118.6 (C) 11.86 (D) 1186

Open solution

Q91. JEE Main 2016 (Online) 9th April Morning Slot

The amount of arsenic pentasulphide that can be obtained when 35.5 g arsenic acid is treated with excess H_2S in the presence of conc. HCl (assuming 100% conversion) is :

(A) 0.50 mol (B) 0.25 mol (C) 0.125 mol (D) 0.333 mol

Open solution

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Q92. JEE Main 2016 (Online) 9th April Morning Slot

An organic compound contains C, H and S. The minimum molecular weight of the compound containing 8% sulphur is :

(atomic weight of S = 32 amu)

Open solution

(A) 200 g mol^{-1} (B) 400 g mol^{-1} (C) 600 g mol^{-1} (D) 300 g mol^{-1}

Q93. JEE Main 2016 (Offline)

At 300 K and 1 atm, 15 mL of a gaseous hydrocarbon requires 375 mL air containing 20% O_2 by volume for complete combustion. After combustion the gases occupy 345 mL. Assuming that the water formed is in liquid form and the volumes were measured at the same temperature and pressure, the formula of the hydrocarbon is:

Open solution

(A) C_3H_8 (B) C_4H_8 (C) C_4H_{10} (D) C_3H_6

Q94. JEE Main 2015 (Online) 11th April Morning Slot

$A + 2B + 3C \rightleftharpoons AB_2C_3$

Reaction of 6.0 g of A, 6.0×10^{23} atoms of B, and 0.036 mol of C yields 4.8 g of compound AB_2C_3. If the atomic mass of A and C are 60 and 80 amu, respectively, the atomic mass of B is (Avogadro no. = 6×10^{23}) :

Open solution

(A) 70 amu (B) 60 amu (C) 50 amu (D) 40 amu

Q95. JEE Main 2015 (Online) 10th April Morning Slot

A sample of a hydrate of barium chloride weighing 61 g was heated until all the water of hydration is removed. The dried sample weighed 52 g. The formula of the hydrated salt is : (atomic mass, Ba = 137 amu, Cl = 35.5 amu)

Open solution

(A) $BaCl_2 . H_2O$ (B) $BaCl_2 . 2H_2O$ (C) $BaCl_2 . 3H_2O$ (D) $BaCl_2 . 4H_2O$

Q96. JEE Main 2015 (Offline)

The molecular formula of a commercial resin used for exchanging ions in water softening is $C_8H_7SO_3Na$ (Mol. Wt. 206). What would be the maximum uptake of Ca^{2+} ions by the resin when expressed in mole per gram resin?

Open solution

(A) 2/309 (B) 1/412 (C) 1/103 (D) 1/206

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Q97. JEE Main 2014 (Online) 12th April Morning Slot

The amount of $BaSO_4$ formed upon mixing 100 mL of 20.8% $BaCl_2$ solution with 50 mL of 9.8% H_2SO_4 solution will be :

(Ba = 137, Cl = 35.5, S = 32, H = 1and O = 16)

(A) 23.3 g (B) 11.65 g (C) 30.6 g (D) 33.2 g

Open solution

Q98. JEE Main 2014 (Online) 11th April Morning Slot

A gaseous compound of nitrogen and hydrogen contains 12.5% (by mass) of hydrogen. The density of the compound relative to hydrogen is 16. The molecular formula of the compound is :

(A) NH_2 (B) N_3H (C) NH_3 (D) N_2H_4

Open solution

Q99. JEE Main 2014 (Online) 9th April Morning Slot

Dissolving 120 g of a compound of (mol. wt. 60) in 1000 g of water gave a solution of density 1.12 g/mL. The molarity of the solution is :

(A) 1.00 M (B) 2.00 M (C) 2.50 M (D) 4.00 M

Open solution

Q100. JEE Main 2014 (Online) 9th April Morning Slot

The amount of oxygen in 3.6 moles of water is :

(A) 115.2 g (B) 57.6 g (C) 28.8 g (D) 18.4 g

Open solution

Q101. JEE Main 2014 (Offline)

The ratio of masses of oxygen and nitrogen in a particular gaseous mixture is 1 : 4. The ratio of number of their molecules is:

(A) 1: 8 (B) 3 : 16 (C) 1 : 4 (D) 7 : 32

Open solution

Q102. JEE Main 2013 (Online) 25th April Morning Slot

6 litres of an alkene require 27 litres of oxygen at constant temperature and pressure for complete combustion. The alkene is :

(A) Ethene (B) Propene (C) 1-Butene (D) 2-Butene

Open solution

Q103. JEE Main 2013 (Online) 25th April Morning Slot

10 mL of 2(M) NaOH solution is added to 200 mL of 0.5(M) of NaOH solution. What is the final concentration?

(A) 0.57(M) (B) 5.7(M) (C) 11.4(M) (D) 1.14(M)

Open solution

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Q104. JEE Main 2013 (Online) 23th April Morning Slot

Number of atoms in the following samples of substances is largest in:

(A) 4.0 g of hydrogen
(B) 71.0 g of chlorine
(C) 127.0 g of iodine
(D) 48.0 g of magnesium

Open solution

Q105. JEE Main 2013 (Online) 22th April Morning Slot

The density of 3M solution of sodium chloride is 1.252 g mL^{-1}. The molality of the solution will be : (molar mass, NaCl = 58.5 g mol^{-1})

(A) 2.60 m
(B) 2.18 m
(C) 2.79 m
(D) 3.00 m

Open solution

Q106. JEE Main 2013 (Offline)

A gaseous hydrocarbon gives upon combustion 0.72 g of water and 3.08 g of CO_2. The empirical formula of the hydrocarbon is

(A) C_2H_4
(B) C_3H_4
(C) C_6H_5
(D) C_7H_8

Open solution

Q107. JEE Main 2013 (Offline)

Experimentally it was found that a metal oxide has formula $M_{0.98}O$. Metal M, present as M^{2+} and M^{3+} in its oxide. Fraction of the metal which exists as M^{3+} would be

(A) 7.01%
(B) 4.08%
(C) 6.05%
(D) 5.08%

Open solution

Q108. JEE Main 2013 (Offline)

The molarity of a solution obtained by mixing 750 mL of 0.5 (M) HCl with 250 mL of 2(M) HCl will be:

(A) 1.00 M
(B) 1. 75 M
(C) 0.975 M
(D) 0.875 M

Open solution

Q109. AIEEE 2012 (Online) 26th May

A transition metal M forms a volatile chloride which has a vapour density of 94.8. If it contains 74.75% of chlorine the formula of the metal chloride will be

(A) MCl_3
(B) MCl_2
(C) MCl_4
(D) MCl_5

Open solution

Q110. AIEEE 2012 (Online) 19th May

When $CO_2(g)$ is passed over red hot coke it partially gets reduced to $CO(g)$. Upon passing 0.5 L of $CO_2(g)$ over red hot coke, the total volume of the gases increased to 700 mL. The composition of the gaseous mixture at STP is

(A) $CO_2 = 300$ mL; $CO = 400$ mL
(B) $CO_2 = 0.0$ mL; $CO = 700$ mL
(C) $CO_2 = 200$ mL; $CO = 500$ mL
(D) $CO_2 = 350$ mL; $CO = 350$ mL

Open solution

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Q111. AIEEE 2012 (Online) 12th May

An aqueous solution of oxalic acid dihydrate contains its $6.3\ \text{g}$ in $250\ \text{mL}$. The volume of $0.1\ \text{N}\ \text{NaOH}$ required to completely neutralize $10\ \text{mL}$ of this solution

(A) $4\ \text{mL}$ (B) $20\ \text{mL}$ (C) $2\ \text{mL}$ (D) $40\ \text{mL}$

Open solution

Q112. AIEEE 2012 (Online) 12th May

$5\ \text{g}$ of benzene on nitration gave $6.6\ \text{g}$ of nitrobenzene. The theoretical yield of the nitrobenzene will be

(A) $4.5\ \text{g}$ (B) $5.6\ \text{g}$ (C) $8.09\ \text{g}$ (D) $6.6\ \text{g}$

Open solution

Q113. AIEEE 2012 (Online) 12th May

The ppm level of F^- in a $500\ \text{g}$ sample of a tooth paste containing $0.2\ \text{g}\ \text{F}^-$ is

(A) 400 (B) 1000 (C) 250 (D) 200

Open solution

Q114. AIEEE 2012 (Online) 7th May

The concentrated sulphuric acid that is peddled commercially is $95\%\ \text{H}_2\text{SO}_4$ by weight. If the density of this commercial acid is $1.834\ \text{g}\,\text{cm}^{-3}$, the molarity of this solution is

(A) $17.8\ \text{M}$ (B) $12.0\ \text{M}$ (C) $10.5\ \text{M}$ (D) $15.7\ \text{M}$

Open solution

Q115. AIEEE 2012 (Online) 7th May

The ratio of number of oxygen atoms (O) in $16.0\ \text{g}$ ozone (O_3), $28.0\ \text{g}$ carbon monoxide (CO) and $16.0\ \text{g}$ oxygen (O_2) is (Atomic mass : $\text{C} = 12$, $\text{O} = 16$ and Avogadro's constant $N_\text{A} = 6.0 \times 10^{23}\ \text{mol}^{-1}$)

(A) $3 : 1 : 2$ (B) $1 : 1 : 2$ (C) $3 : 1 : 1$ (D) $1 : 1 : 1$

Open solution

Q116. AIEEE 2012

The density of a solution prepared by dissolving 120 g of urea (mol. Mass = 60 u) in 1000g of water is 1.15 g/mL. The molarity of this solution is :

(A) 0.50 M (B) 1.78 M (C) 1.02 M (D) 2.05 M

Open solution

Q117. AIEEE 2011

A 5.2 molal aqueous solution of methyl alcohol, CH_3OH, is supplied. What is the mole fraction of methyl alcohol in the solution ?

(A) 0.190 (B) 0.086 (C) 0.050 (D) 0.100

Open solution

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Q118. AIEEE 2007

The density (in g mL^{-1}) of a 3.60 M sulphuric acid solution that is 29% H_2SO_4 (molar mass = 98 g mol^{-1}) by mass will be

(A) 1.45 (B) 1.64 (C) 1.88 (D) 1.22

Open solution

Q119. AIEEE 2007

In the reaction

$2Al(s) + 6HCl(aq) \rightarrow 2Al^{3+}(aq) + 6Cl^{-}(aq) + 3H_2(g)$

(A) 11.2 L $H_2(g)$ at STP is produced for every mole HCl(aq) consumed

(B) 6 L HCl(aq) is consumed for every 3L $H_2(g)$ produced

(C) 33.6 L $H_2(g)$ is produced regardless of temperature and pressure for every mole that reacts

(D) 67.2 $H_2(g)$ at STP is produced for every mole Al that reacts

Open solution

Q120. AIEEE 2006

How many moles of magnesium phosphate $Mg_3(PO_4)_2$ will contain 0.25 mole of oxygen atoms?

(A) 1.25×10^{-2} (B) 2.5×10^{-2} (C) 0.02 (D) 3.125×10^{-2}

Open solution

Q121. AIEEE 2006

Density of a 2.05M solution of acetic acid in water is 1.02 g/mL. The molality of the solution is

(A) 2.28 mol kg^{-1} (B) 0.44 mol kg^{-1} (C) 1.14 mol kg^{-1} (D) 3.28 mol kg^{-1}

Open solution

Q122. AIEEE 2005

Two solutions of a substance (non electrolyte) are mixed in the following manner. 480 ml of 1.5 M first solution + 520 ml of 1.2 M second solution. What is the molarity of the final mixture?

(A) 2.70 M (B) 1.344 M (C) 1.50 M (D) 1.20 M

Open solution

Q123. AIEEE 2005

If we consider that 1/6, in place of 1/12, mass of carbon atom is taken to be the relative atomic mass unit, the mass of one mole of the substance will

(A) be a function of the molecular mass of the substance

(B) remain unchanged

(C) increase two fold

(D) decrease twice

Open solution

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Q124. AIEEE 2004

6.02×10^{20} molecules of urea are present in 100 ml of its solution. The concentration of urea solution is

(Avogadro constant, $N_A = 6.02 \times 10^{23}$ mol^{-1})

(A) 0.02 M (B) 0.01 M (C) 0.001 M (D) 0.1 M

Open solution

Q125. AIEEE 2004

To neutralise completely 20 mL of 0.1 M aqueous solution of phosphorous acid (H_3PO_3), the volume of 0.1 M aqueous KOH solution required is

(A) 40 mL (B) 20 mL (C) 10 mL (D) 60 mL

Open solution

Q126. AIEEE 2003

What volume of hydrogen gas at 273 K and 1 atm pressure will be consumed in obtaining 21.6 g of elemental boron (atomic mass = 10.8) from the reduction of boron trichloride by hydrogen?

(A) 67.2 L (B) 44.8 L (C) 22.4 L (D) 89.6 L

Open solution

Q127. AIEEE 2003

25 ml of a solution of barium hydroxide on titration with a 0.1 molar solution of hydrochloric acid gave a titre value of 35 ml. The molarity of barium hydroxide solution was

(A) 0.14 (B) 0.28 (C) 0.35 (D) 0.07

Open solution

Q128. AIEEE 2002

In a compound C, H and N atoms are present in 9 : 1 : 3.5 by weight . Molecular weight of the compound is 108 g mol^{-1} . Molecular formula of compound is

(A) $C_2H_6N_2$ (B) C_3H_4N (C) $C_6H_8N_2$ (D) $C_9H_{12}N_3$

Open solution

Q129. AIEEE 2002

With increase of temperature, which of these changes?

(A) molality (B) weight fraction of solute

(C) molarity (D) mole fraction

Open solution

Q130. AIEEE 2002

Number of atoms in 558.5 gram Fe (at. wt. of Fe = 55.85 g mol^{-1}) is

(A) twice that in 60 g carbon (B) 6.023×10^{22}

(C) half that in 8 g He (D) $558.5 \times 6.023 \times 10^{23}$

Open solution

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Numerical 124 questions

Q131. JEE Main 2026 (Online) 24th January Evening Shift

0.25 g of an organic compound "A" containing carbon, hydrogen and oxygen was analysed using the combustion method. There was an increase in mass of $CaCl_2$ tube and potash tube at the end of the experiment. The amount was found to be 0.15 g and 0.1837 g , respectively. The percentage of oxygen in compound A is

____ %. (Nearest integer)

(Given: molar mass in $gmol^{-1}$ $H: 1, C: 12, O: 16$)

Open solution

Q132. JEE Main 2026 (Online) 23rd January Morning Shift

x mg of pure HCl was used to make an aqueous solution. 25.0 mL of $0.1\ M\ Ba(OH)_2$ solution is used when the HCl solution was titrated against it. The numerical value of x is ____ $\times 10^{-1}$. (Nearest integer)

Given : Molar mass of HCl and $Ba(OH)_2$ are 36.5 and $171.0\ g\ mol^{-1}$ respectively.

Open solution

Q133. JEE Main 2025 (Online) 8th April Evening Shift

20 mL of sodium iodide solution gave 4.74 g silver iodide when treated with excess of silver nitrate solution. The molarity of the sodium iodide solution is _____ M. (Nearest Integer value)

(Given : Na = 23, I = 127, Ag = 108, N = 14, O = 16 g mol^{-1})

Open solution

Q134. JEE Main 2025 (Online) 7th April Evening Shift

Butane reacts with oxygen to produce carbon dioxide and water following the equation given below.

$C_4H_{10}(g) + \frac{13}{2}O_2(g) \rightarrow 4CO_2(g) + 5H_2O(l)$

If 174.0 kg of butane is mixed with 320.0 kg of O_2, the volume of water formed in liters is

_____. (Nearest integer)

[Given : (a) Molar mass of C, H, O are $12, 1, 16\ g\ mol^{-1}$ respectively, (b) Density of water $= 1\ g\ mL^{-1}$]

Open solution

Q135. JEE Main 2025 (Online) 7th April Morning Shift

An organic compound weighing 500 mg , produced 220 mg of CO_2, on complete combustion. The percentage composition of carbon in the compound is _____ %. (nearest integer)

(Given molar mass in $gmol^{-1}$ of $C: 12, O: 16$)

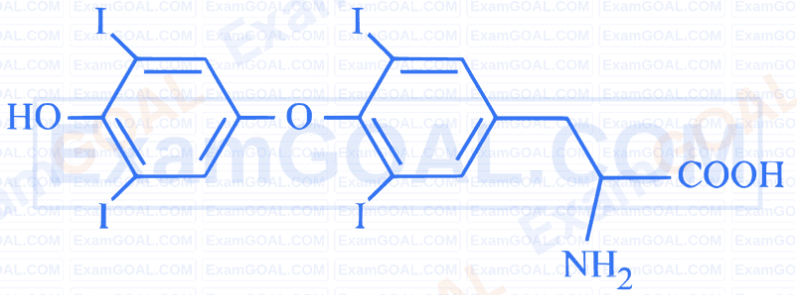
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Q136. JEE Main 2025 (Online) 7th April Morning Shift

Open solution

Thyroxine, the hormone has given below structure



The percentage of iodine in thyroxine is _____ %. (nearest integer)

(Given molar mass in gmol^{-1} $\mathrm{C}: 12, \mathrm{H}: 1, \mathrm{O}: 16,\ \mathrm{N}: 14, \mathrm{I}: 127$)

Q137. JEE Main 2025 (Online) 4th April Evening Shift

Open solution

Sea water, which can be considered as a 6 molar $(6\mathrm{M})$ solution of NaCl , has a density of $2\ \mathrm{g\ mL}^{-1}$. The concentration of dissolved oxygen (O_2) in sea water is 5.8 ppm . Then the concentration of dissolved oxygen (O_2) in sea water, in terms of molality, is $x \times 10^{-4}$ m.

$x =$ _____. (Nearest integer)

Given: Molar mass of NaCl is $58.5\ \mathrm{g\ mol}^{-1}$

Molar mass of O_2 is $32\ \mathrm{g\ mol}^{-1}$

Q138. JEE Main 2025 (Online) 4th April Evening Shift

Open solution

The amount of calcium oxide produced on heating 150 kg limestone (75% pure) is _____ kg. (Nearest integer)

Given: Molar mass (in gmol^{-1}) of Ca-40, O-16, C-12

Q139. JEE Main 2025 (Online) 4th April Morning Shift

Open solution

Fortification of food with iron is done using $\mathrm{FeSO_4 \cdot 7H_2O}$. The mass in grams of the $\mathrm{FeSO_4 \cdot 7H_2O}$ required to achieve 12 ppm of iron in 150 kg of wheat is _____ (Nearest integer)

[Given: Molar mass of Fe, S and O respectively are 56, 32 and $16\ \mathrm{g\ mol}^{-1}$]

Q140. JEE Main 2025 (Online) 3rd April Evening Shift

Open solution

X g of nitrobenzene on nitration gave 4.2 g of m -dinitrobenzene. X = _____g. (nearest integer)

[Given : molar mass (in gmol^{-1}) $\mathrm{C}: 12, \mathrm{H}: 1, \mathrm{O}: 16,\ \mathrm{N}: 14$]

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Q141. JEE Main 2025 (Online) 3rd April Morning Shift

0.5 g of an organic compound on combustion gave 1.46 g of CO_2 and 0.9 g of H_2O. The percentage of carbon in the compound is _____. (Nearest integer)

[Given : Molar mass (in $gmol^{-1}$) $C : 12, H : 1, O : 16$]

Open solution

Q142. JEE Main 2025 (Online) 28th January Morning Shift

The molarity of a 70% (mass/mass) aqueous solution of a monobasic acid (X) is _____ $\times 10^{-1}$ M (Nearest integer)

[Given: Density of aqueous solution of (X) is $1.25\ g\ mL^{-1}$

Molar mass of the acid is $70\ g\ mol^{-1}$]

Open solution

Q143. JEE Main 2025 (Online) 28th January Morning Shift

Quantitative analysis of an organic compound (X) shows following % composition.

C : 14.5%

Cl : 64.46%

H: 1.8 %

Empirical formula mass of the compound (X) is _____ $\times 10^{-1}$

(Given molar mass in $g\ mol^{-1}$ of $C : 12, H : 1, O : 16, Cl : 35.5$)

Open solution

Q144. JEE Main 2025 (Online) 24th January Morning Shift

Consider the following reaction occurring in the blast furnace:

$Fe_3O_{4(s)} + 4CO_{(g)} \rightarrow 3Fe_{(l)} + 4CO_{2(g)}$

' x ' kg of iron is produced when $2.32 \times 10^3\ kg Fe_3O_4$ and $2.8 \times 10^2\ kg CO$ are brought together in the furnace. The value of ' x ' is _____ . (nearest integer)

{Given: molar mass of $Fe_3O_4 = 232\ g\ mol^{-1}$

molar mass of $CO = 28\ g\ mol^{-1}$

molar mass of $Fe = 56\ g\ mol^{-1}$}

Open solution

Q145. JEE Main 2025 (Online) 24th January Morning Shift

Xg of benzoic acid on reaction with aq $NaHCO_3$ released CO_2 that occupied 11.2 L volume at STP.

X is _____ g.

Open solution

Q146. JEE Main 2025 (Online) 23rd January Evening Shift

0.01 mole of an organic compound (X) containing 10% hydrogen, on complete combustion produced $0.9\ g H_2O$. Molar mass of (X) is _____ $g\ mol^{-1}$.

Open solution

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Q147. JEE Main 2025 (Online) 23rd January Evening Shift

When 81.0 g of aluminium is allowed to react with 128.0 g of oxygen gas, the mass of aluminium oxide produced in grams is _____ . (Nearest integer)

Given :

Molar mass of Al is $27.0\ \mathrm{g\ mol^{-1}}$

Molar mass of O is $16.0\ \mathrm{g\ mol^{-1}}$

Open solution

Q148. JEE Main 2025 (Online) 22nd January Evening Shift

20 mL of 2 M NaOH solution is added to 400 mL of 0.5 M NaOH solution. The final concentration of the solution is _____ $\times 10^{-2}\mathrm{M}$. (Nearest integer)

Open solution

Q149. JEE Main 2024 (Online) 9th April Morning Shift

Molarity (M) of an aqueous solution containing x g of anhyd. $\mathrm{CuSO_4}$ in $500\ \mathrm{mL}$ solution at $32^{\circ}\mathrm{C}$ is $2 \times 10^{-1}\mathrm{M}$. Its molality will be _____ $\times 10^{-3}$ m. (nearest integer). [Given density of the solution $= 1.25\ \mathrm{g/mL}$]

Open solution

Q150. JEE Main 2024 (Online) 8th April Evening Shift

A solution is prepared by adding 1 mole ethyl alcohol in 9 mole water. The mass percent of solute in the solution is _____ (Integer answer) (Given : Molar mass in $\mathrm{g\ mol^{-1}}$ Ethyl alcohol : 46 water : 18)

Open solution

Q151. JEE Main 2024 (Online) 8th April Evening Shift

Molality of an aqueous solution of urea is 4.44 m. Mole fraction of urea in solution is $x \times 10^{-3}$, Value of x is _____. (Integer answer)

Open solution

Q152. JEE Main 2024 (Online) 4th April Evening Shift

From 6.55 g of aniline, the maximum amount of acetanilide that can be prepared will be _____ $\times 10^{-1}$ g.

Open solution

Q153. JEE Main 2024 (Online) 4th April Morning Shift

$\mathrm{X g}$ of ethylamine is subjected to reaction with $\mathrm{NaNO_2/HCl}$ followed by water; evolved dinitrogen gas which occupied $2.24\ \mathrm{L}$ volume at STP. X is _____ $\times 10^{-1}$ g.

Open solution

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Q154. JEE Main 2024 (Online) 1st February Evening Shift

$10\ \mathrm{mL}$ of gaseous hydrocarbon on combustion gives $40\ \mathrm{mL}$ of $\mathrm{CO_2(g)}$ and $50\ \mathrm{mL}$ of water vapour. Total number of carbon and hydrogen atoms in the hydrocarbon is _____ .

Open solution

Q155. JEE Main 2024 (Online) 1st February Morning Shift

Consider the following reaction :

$$3\mathrm{PbCl_2} + 2(\mathrm{NH_4})_3\mathrm{PO_4} \rightarrow \mathrm{Pb_3(PO_4)_2} + 6\mathrm{NH_4Cl}$$

If $72\ \mathrm{mmol}$ of $\mathrm{PbCl_2}$ is mixed with $50\ \mathrm{mmol}$ of $(\mathrm{NH_4})_3\mathrm{PO_4}$, then the amount of $\mathrm{Pb_3(PO_4)_2}$ formed is _____ mmol (nearest integer).

Open solution

Q156. JEE Main 2024 (Online) 31st January Evening Shift

The molarity of $1\ \mathrm{L}$ orthophosphoric acid $(\mathrm{H_3PO_4})$ having 70% purity by weight (specific gravity $1.54\ \mathrm{g\ cm^{-3}}$) is _____ M.

(Molar mass of $\mathrm{H_3PO_4} = 98\ \mathrm{g\ mol^{-1}}$)

Open solution

Q157. JEE Main 2024 (Online) 31st January Morning Shift

Number of moles of methane required to produce $22\ \mathrm{g}\ \mathrm{CO_{2(g)}}$ after combustion is $\mathrm{x} \times 10^{-2}$ moles. The value of x is _____.

Open solution

Q158. JEE Main 2024 (Online) 31st January Morning Shift

Molar mass of the salt from $\mathrm{NaBr}, \mathrm{NaNO_3}, \mathrm{KI}$ and $\mathrm{CaF_2}$ which does not evolve coloured vapours on heating with concentrated $\mathrm{H_2SO_4}$ is _____ $\mathrm{g\ mol^{-1}}$.

(Molar mass in $\mathrm{g\ mol^{-1}}$: $\mathrm{Na} : 23,\ \mathrm{N} : 14,\ \mathrm{K} : 39, \mathrm{O} : 16, \mathrm{Br} : 80, \mathrm{I} : 127,\ \mathrm{F} : 19, \mathrm{Ca} : 40$)

Open solution

Q159. JEE Main 2024 (Online) 30th January Morning Shift

The mass of sodium acetate $(\mathrm{CH_3COONa})$ required to prepare $250\ \mathrm{mL}$ of $0.35\ \mathrm{M}$ aqueous solution is _____ g. (Molar mass of $\mathrm{CH_3COONa}$ is $82.02\ \mathrm{g\ mol^{-1}}$)

Open solution

Q160. JEE Main 2024 (Online) 30th January Morning Shift

$0.05\ \mathrm{cm}$ thick coating of silver is deposited on a plate of $0.05\ \mathrm{m^2}$ area. The number of silver atoms deposited on plate are _____ $\times 10^{23}$. (At mass $\mathrm{Ag} = 108,\ \mathrm{d} = 7.9\ \mathrm{g\ cm^{-3}}$)

Open solution

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Q161. JEE Main 2024 (Online) 29th January Evening Shift

If $50\ \mathrm{mL}$ of $0.5\mathrm{M}$ oxalic acid is required to neutralise $25\ \mathrm{mL}$ of NaOH solution, the amount of NaOH in $50\ \mathrm{mL}$ of given NaOH solution is _____ g.

Open solution

Q162. JEE Main 2024 (Online) 29th January Evening Shift

Molality of 0.8 M H_2SO_4 solution (density 1.06 g cm^{-3}) is _____ $\times 10^{-3}$ m.

Open solution

Q163. JEE Main 2024 (Online) 29th January Morning Shift

A solution of $\mathrm{H_2SO_4}$ is $31.4\%\mathrm{H_2SO_4}$ by mass and has a density of $1.25\ \mathrm{g/mL}$. The molarity of the $\mathrm{H_2SO_4}$ solution is _____ M (nearest integer)

[Given molar mass of $\mathrm{H_2SO_4} = 98\ \mathrm{g\ mol^{-1}}$]

Open solution

Q164. JEE Main 2024 (Online) 27th January Evening Shift

$9.3\ \mathrm{g}$ of aniline is subjected to reaction with excess of acetic anhydride to prepare acetanilide. The mass of acetanilide produced if the reaction is 100% completed is _____ $\times 10^{-1}\ \mathrm{g}$.

(Given molar mass in $\mathrm{g\ mol^{-1}}$

$\mathrm{N}: 14, \mathrm{O}: 16,$
$\mathrm{C}: 12, \mathrm{H}: 1$)

Open solution

Q165. JEE Main 2024 (Online) 27th January Evening Shift

Volume of $3\mathrm{M}\ \mathrm{NaOH}$ (formula weight $40\ \mathrm{g\ mol^{-1}}$) which can be prepared from $84\ \mathrm{g}$ of NaOH is _____ $\times 10^{-1}\mathrm{dm}^3$.

Open solution

Q166. JEE Main 2024 (Online) 27th January Morning Shift

Mass of methane required to produce $22\ \mathrm{g}$ of $\mathrm{CO_2}$ after complete combustion is _____ g.

(Given Molar mass in g mol-1 $\mathrm{C} = 12.0, \mathrm{H} = 1.0, \mathrm{O} = 16.0$)

Open solution

Q167. JEE Main 2023 (Online) 15th April Morning Shift

The volume (in mL) of $0.1\mathrm{M}\ \mathrm{AgNO_3}$ required for complete precipitation of chloride ions present in $20\ \mathrm{mL}$ of $0.01\mathrm{M}$ solution of $[\mathrm{Cr(H_2O)_5Cl}]\mathrm{Cl_2}$ as silver chloride is _____.

Open solution

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Q168. JEE Main 2023 (Online) 13th April Evening Shift

$1\ \mathrm{g}$ of a carbonate $(\mathrm{M_2CO_3})$ on treatment with excess HCl produces $0.01\ \mathrm{mol}$ of $\mathrm{CO_2}$. The molar mass of $\mathrm{M_2CO_3}$ is _____ $\mathrm{g\ mol^{-1}}$. (Nearest integer)

Open solution

Q169. JEE Main 2023 (Online) 13th April Morning Shift

An organic compound gives $0.220\ \mathrm{g}$ of $\mathrm{CO_2}$ and $0.126\ \mathrm{g}$ of $\mathrm{H_2O}$ on complete combustion. If the $\%$ of carbon is 24 then the $\%$ of hydrogen is _____ $\times 10^{-1}$. (Nearest integer)

Open solution

Q170. JEE Main 2023 (Online) 13th April Morning Shift

$20\ \mathrm{mL}$ of calcium hydroxide was consumed when it was reacted with $10\ \mathrm{mL}$ of unknown solution of $\mathrm{H_2SO_4}$. Also $20\ \mathrm{mL}$ standard solution of $0.5\ \mathrm{M\ HCl}$ containing 2 drops of phenolphthalein was titrated with calcium hydroxide, the mixture showed pink colour when burette displayed the value of $35.5\ \mathrm{mL}$ whereas the burette showed $25.5\ \mathrm{mL}$ initially. The concentration of $\mathrm{H_2SO_4}$ is _____ M. (Nearest integer)

Open solution

Q171. JEE Main 2023 (Online) 11th April Evening Shift

The volume of hydrogen liberated at STP by treating $2.4\ \mathrm{g}$ of magnesium with excess of hydrochloric acid is _____ $\times\ 10^{-2}\ \mathrm{L}$

Given : Molar volume of gas is $22.4\ \mathrm{L}$ at STP.

Molar mass of magnesium is $24\ \mathrm{g\ mol^{-1}}$

Open solution

Q172. JEE Main 2023 (Online) 11th April Morning Shift

A solution of sugar is obtained by mixing $200\ \mathrm{g}$ of its 25% solution and $500\ \mathrm{g}$ of its 40% solution (both by mass). The mass percentage of the resulting sugar solution is _____ (Nearest integer)

Open solution

Q173. JEE Main 2023 (Online) 8th April Morning Shift

$0.5\ \mathrm{g}$ of an organic compound (X) with 60% carbon will produce _____ $\times 10^{-1}\ \mathrm{g}$ of $\mathrm{CO_2}$ on complete combustion.

Open solution

Q174. JEE Main 2023 (Online) 6th April Morning Shift

If 5 moles of $\mathrm{BaCl_2}$ is mixed with 2 moles of $\mathrm{Na_3PO_4}$, the maximum number of moles of $\mathrm{Ba_3(PO_4)_2}$ formed is _____ (Nearest integer)

Open solution

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Q175. JEE Main 2023 (Online) 1st February Evening Shift

The molality of a $10\%(\mathrm{v/v})$ solution of di-bromine solution in $\mathrm{CCl_4}$ (carbon tetrachloride) is 'x'. $x =$ _____ $\times 10^{-2}$ m. (Nearest integer)

[Given : molar mass of $\mathrm{Br_2} = 160\ \mathrm{g\ mol^{-1}}$

atomic mass of $\mathrm{C} = 12\ \mathrm{g\ mol^{-1}}$

atomic mass of $\mathrm{Cl} = 35.5\ \mathrm{g\ mol^{-1}}$

density of dibromine $= 3.2\ \mathrm{g\ cm^{-3}}$

density of $\mathrm{CCl_4} = 1.6\ \mathrm{g\ cm^{-3}}$]

Open solution

Q176. JEE Main 2023 (Online) 1st February Morning Shift

The density of $3\mathrm{M}$ solution of NaCl is $1.0\ \mathrm{g\ mL^{-1}}$. Molality of the solution is _____ $\times 10^{-2}$ m. (Nearest integer).

Given: Molar mass of Na and Cl is 23 and $35.5\ \mathrm{g\ mol^{-1}}$ respectively.

Open solution

Q177. JEE Main 2023 (Online) 31st January Evening Shift

Assume carbon burns according to the following equation :

$$2\mathrm{C_{(s)}} + \mathrm{O_{2(g)}} \rightarrow 2\mathrm{CO(g)}$$

when $12\ \mathrm{g}$ carbon is burnt in $48\ \mathrm{g}$ of oxygen, the volume of carbon monoxide produced is _____ $\times 10^{-1}\ \mathrm{L}$ at STP [nearest integer]

[Given: Assume CO as ideal gas, Mass of C is $12\ \mathrm{g\ mol^{-1}}$, Mass of O is $16\ \mathrm{g\ mol^{-1}}$ and molar volume of an ideal gas at STP is $22.7\ \mathrm{L\ mol^{-1}}$]

Open solution

Q178. JEE Main 2023 (Online) 31st January Evening Shift

A sample of a metal oxide has formula $\mathrm{M_{0.83}O_{1.00}}$. The metal M can exist in two oxidation states $+2$ and $+3$.

In the sample of $\mathrm{M_{0.83}O_{1.00}}$, the percentage of metal ions existing in $+2$ oxidation state is _____ %. (nearest integer)

Open solution

Q179. JEE Main 2023 (Online) 31st January Morning Shift

Zinc reacts with hydrochloric acid to give hydrogen and zinc chloride. The volume of hydrogen gas produced at STP from the reaction of $11.5\ \mathrm{g}$ of zinc with excess HCl is _____ L (Nearest integer)

(Given : Molar mass of Zn is $65.4\ \mathrm{g\ mol^{-1}}$ and Molar volume of $\mathrm{H_2}$ at $\mathrm{STP} = 22.7\ \mathrm{L}$)

Open solution

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Q180. JEE Main 2023 (Online) 31st January Morning Shift

On complete combustion, 0.492 g of an organic compound gave 0.792 g of CO_2. The % of carbon in the organic compound is _____ (Nearest integer)

Open solution

Q181. JEE Main 2023 (Online) 30th January Evening Shift

The strength of 50 volume solution of hydrogen peroxide is _____ g/L (Nearest integer).

Given:

Molar mass of H_2O_2 is $34\ g\ mol^{-1}$

Molar volume of gas at $STP = 22.7\ L$

Open solution

Q182. JEE Main 2023 (Online) 30th January Morning Shift

Some amount of dichloromethane (CH_2Cl_2) is added to $671.141\ mL$ of chloroform $(CHCl_3)$ to prepare $2.6 \times 10^{-3}M$ solution of $CH_2Cl_2(DCM)$. The concentration of DCM is _____ ppm (by mass).

Given :

atomic mass : C = 12

H = 1

Cl = 35.5

density of $CHCl_3 = 1.49\ g\ cm^{-3}$

Open solution

Q183. JEE Main 2023 (Online) 29th January Evening Shift

The volume of HCl, containing 73 g L^{-1}, required to completely neutralise NaOH obtained by reacting 0.69 g of metallic sodium with water, is _____ mL. (Nearest Integer)

(Given : molar masses of Na, Cl, O, H, are 23, 35.5, 16 and 1 g mol^{-1} respectively.)

Open solution

Q184. JEE Main 2023 (Online) 29th January Evening Shift

When 0.01 mol of an organic compound containing 60% carbon was burnt completely, 4.4 g of CO_2 was produced. The molar mass of compound is _____ g mol^{-1} (Nearest integer).

Open solution

Q185. JEE Main 2023 (Online) 25th January Evening Shift

Number of hydrogen atoms per molecule of a hydrocarbon A having 85.8% carbon is _____

(Given : Molar mass of A = 84 g mol^{-1})

Open solution

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Q186. JEE Main 2023 (Online) 25th January Morning Shift

The density of a monobasic strong acid (Molar mass 24.2 g/mol) is 1.21 kg/L. The volume of its solution required for the complete neutralization of 25 mL of 0.24 M NaOH is _____ $\times 10^{-2}$ mL (Nearest integer)

Open solution

Q187. JEE Main 2023 (Online) 24th January Evening Shift

The number of units, which are used to express concentration of solutions from the following is _____

Mass percent, Mole, Mole fraction, Molarity, ppm, Molality

Open solution

Q188. JEE Main 2023 (Online) 24th January Morning Shift

When $Fe_{0.93}O$ is heated in presence of oxygen, it converts to Fe_2O_3. The number of correct statement/s from the following is _____

A. The equivalent weight of $Fe_{0.93}O$ is $\frac{\text{Molecular weight}}{0.79}$

B. The number of moles of Fe^{2+} and Fe^{3+} in 1 mole of $Fe_{0.93}O$ is 0.79 and 0.14 respectively

C. $Fe_{0.93}O$ is metal deficient with lattice comprising of cubic closed packed arrangement of O^{2-} ions

D. The % composition of Fe^{2+} and Fe^{3+} in $Fe_{0.93}O$ is 85% and 15% respectively

Open solution

Q189. JEE Main 2023 (Online) 24th January Morning Shift

5 g of NaOH was dissolved in deionized water to prepare a 450 mL stock solution. What volume (in mL) of this solution would be required to prepare 500 mL of 0.1 M solution? _____

Given : Molar Mass of Na, O and H is 23, 16 and 1 g mol^{-1} respectively

Open solution

Q190. JEE Main 2022 (Online) 30th July Evening Shift

The density of 4 molal solution of NaOH is $1.160\ g\ mL^{-1}$. The molarity of the solution is _____ M. (Given : Molar mass of NaOH $= 40\ g\ mol^{-1}$) (nearest integer)

Open solution

Q191. JEE Main 2022 (Online) 29th July Evening Shift

A 1.84 mg sample of polyhydric alcoholic compound 'X' of molar mass 92.0 g/mol gave 1.344 mL of H_2 gas at STP. The number of alcoholic hydrogens present in compound 'X' is _____.

Open solution

Q192. JEE Main 2022 (Online) 28th July Evening Shift

2L of 0.2M H_2SO_4 is reacted with 2L of 0.1M NaOH solution, the molarity of the resulting product Na_2SO_4 in the solution is _____ millimolar. (Nearest integer)

Open solution

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Q193. JEE Main 2022 (Online) 28th July Morning Shift

In the given reaction,

$X + Y + 3Z \leftrightarrows XYZ_3$

if one mole of each of X and Y with 0.05 mol of Z gives compound XYZ_3. (Given : Atomic masses of X, Y and Z are 10, 20 and 30 amu, respectively.) The yield of XYZ_3 is _____ g. (Nearest integer)

Open solution

Q194. JEE Main 2022 (Online) 28th July Morning Shift

On complete combustion of 0.492 g of an organic compound containing C, H and O, 0.7938 g of CO_2 and 0.4428 g of H_2O was produced. The % composition of oxygen in the compound is _____.

Open solution

Q195. JEE Main 2022 (Online) 27th July Morning Shift

20 mL of 0.02 M $K_2Cr_2O_7$ solution is used for the titration of 10 mL of Fe^{2+} solution in the acidic medium.

The molarity of Fe^{2+} solution is _____ $\times 10^{-2}$ M. (Nearest Integer)

Open solution

Q196. JEE Main 2022 (Online) 26th July Evening Shift

A 100 mL solution of CH_3CH_2MgBr on treatment with methanol produces 2.24 mL of a gas at STP. The weight of gas produced is _____ mg. [nearest integer]

Open solution

Q197. JEE Main 2022 (Online) 26th July Morning Shift

Chlorophyll extracted from the crushed green leaves was dissolved in water to make 2 L solution of Mg of concentration 48 ppm. The number of atoms of Mg in this solution is $x \times 10^{20}$ atoms. The value of x is _____. (Nearest Integer)

(Given : Atomic mass of Mg is 24 g mol^{-1}; $N_A = 6.02 \times 10^{23}$ mol^{-1})

Open solution

Q198. JEE Main 2022 (Online) 26th July Morning Shift

When 800 mL of 0.5 M nitric acid is heated in a beaker, its volume is reduced to half and 11.5 g of nitric acid is evaporated. The molarity of the remaining nitric acid solution is x $\times 10^{-2}$ M. (Nearest integer)

(Molar mass of nitric acid is 63 g mol^{-1})

Open solution

Q199. JEE Main 2022 (Online) 25th July Evening Shift

56.0 L of nitrogen gas is mixed with excess hydrogen gas and it is found that 20 L of ammonia gas is produced. The volume of unused nitrogen gas is found to be _____ L.

Open solution

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Q200. JEE Main 2022 (Online) 25th July Evening Shift

A sample of 4.5 mg of an unknown monohydric alcohol, R-OH was added to methylmagnesium iodide. A gas is evolved and is collected and its volume measured to be 3.1 mL. The molecular weight of the unknown alcohol is _____ g/mol. [Nearest integer]

Open solution

Q201. JEE Main 2022 (Online) 30th June Morning Shift

Blister copper is produced by reaction of copper oxide with copper sulphide.

$2Cu_2O + Cu_2S \rightarrow 6Cu + SO_2$

When 2.86×10^3 g of Cu_2O and 4.77×10^3 g of Cu_2S are used for reaction, the mass of copper produced is _____ g. (nearest integer)

(Atomic mass of Cu = 63.5 a.m. u, S = 32.0 a.m. u, O = 16.0 a.m. u)

Open solution

Q202. JEE Main 2022 (Online) 28th June Evening Shift

The complete combustion of 0.492 g of an organic compound containing 'C', 'H' and 'O' gives 0.793 g of CO_2 and 0.442 g of H_2O. The percentage of oxygen composition in the organic compound is _____. (nearest integer)

Open solution

Q203. JEE Main 2022 (Online) 27th June Evening Shift

116 g of a substance upon dissociation reaction, yields 7.5 g of hydrogen, 60 g of oxygen and 48.5 g of carbon. Given that the atomic masses of H, O and C are 1, 16 and 12, respectively. The data agrees with how many formulae of the following?

A. CH_3COOH, B. HCHO, C. CH_3OOCH_3, D. CH_3CHO

Open solution

Q204. JEE Main 2022 (Online) 27th June Morning Shift

Two elements A and B which form 0.15 moles of A_2B and AB_3 type compounds. If both A_2B and AB_3 weigh equally, then the atomic weight of A is _____ times of atomic weight of B.

Open solution

Q205. JEE Main 2022 (Online) 26th June Evening Shift

CNG is an important transportation fuel. When 100 g CNG is mixed with 208 g oxygen in vehicles, it leads to the formation of CO_2 and H_2O and produces large quantity of heat during this combustion, then the amount of carbon dioxide, produced in grams is _____. [nearest integer]

[Assume CNG to be methane]

Open solution

Q206. JEE Main 2022 (Online) 26th June Evening Shift

The moles of methane required to produce 81 g of water after complete combustion is _____ $\times 10^{-2}$ mol. [nearest integer]

Open solution

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Q207. JEE Main 2022 (Online) 26th June Morning Shift

On complete combustion 0.30 g of an organic compound gave 0.20 g of carbon dioxide and 0.10 g of water. The percentage of carbon in the given organic compound is _____. (Nearest integer)

Open solution

Q208. JEE Main 2022 (Online) 25th June Evening Shift

A protein 'A' contains 0.30% of glycine (molecular weight 75). The minimum molar mass of the protein 'A' is _____ $\times 10^3$ g mol^{-1} [nearest integer]

Open solution

Q209. JEE Main 2022 (Online) 25th June Morning Shift

The number of N atoms in 681 g of $C_7H_5N_3O_6$ is $x \times 10^{21}$. The value of x is ($N_A = 6.02 \times 10^{23}$ mol^{-1}) (Nearest Integer)

Open solution

Q210. JEE Main 2022 (Online) 25th June Morning Shift

1 L aqueous solution of H_2SO_4 contains 0.02 m mol H_2SO_4. 50% of this solution is diluted with deionized water to give 1 L solution (A). In solution (A), 0.01 m mol of H_2SO_4 are added. Total m mols of H_2SO_4 in the final solution is _____ $\times 10^3$ m mols.

Open solution

Q211. JEE Main 2022 (Online) 25th June Morning Shift

Number of grams of bromine that will completely react with 5.0 g of pent-1-ene is _____ $\times 10^{-2}$ g. (Atomic mass of Br = 80 g/mol) [Nearest Integer]

Open solution

Q212. JEE Main 2021 (Online) 1st September Evening Shift

The number of atoms in 8g of sodium is $x \times 10^{23}$. The value of x is _____. (Nearest integer)

[Given : $N_A = 6.02 \times 10^{23}$ mol^{-1}

Atomic mass of Na = 23.0 u]

Open solution

Q213. JEE Main 2021 (Online) 1st September Evening Shift

If 80 g of copper sulphate $CuSO_4 . 5H_2O$ is dissolved in deionised water to make 5L of solution. The concentration of the copper sulphate solution is $x \times 10^{-3}$ mol L^{-1}. The value of x is _____.

[Atomic masses Cu : 63.54u, S : 32u, O : 16u, H : 1u]

Open solution

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Q214. JEE Main 2021 (Online) 31st August Evening Shift

Sodium oxide reacts with water to produce sodium hydroxide. 20.0 g of sodium oxide is dissolved in 500 mL of water. Neglecting the change in volume, the concentration of the resulting NaOH solution is _____ $\times 10^{-1}$ M. (Nearest integer) [Atomic mass : Na = 23.0, O = 16.0, H = 1.0]

Open solution

Q215. JEE Main 2021 (Online) 31st August Morning Shift

The molarity of the solution prepared by dissolving 6.3 g of oxalic acid ($H_2C_2O_4.2H_2O$) in 250 mL of water in mol L^{-1} is x $\times 10^{-2}$. The value of x is _____. (Nearest integer) [Atomic mass : H : 1.0, C : 12.0, O : 16.0]

Open solution

Q216. JEE Main 2021 (Online) 27th August Evening Shift

100 g of propane is completely reacted with 1000 g of oxygen. The mole fraction of carbon dioxide in the resulting mixture is x $\times 10^{-2}$. The value of x is _____. (Nearest integer)

[Atomic weight : H = 1.008; C = 12.00; O = 16.00]

Open solution

Q217. JEE Main 2021 (Online) 26th August Evening Shift

100 mL of Na_3PO_4 solution contains 3.45 g of sodium. The molarity of the solution is _____ $\times 10^{-2}$ mol L^{-1}. (Nearest integer)

[Atomic Masses - Na : 23.0 u, O : 16.0 u, P : 31.0 u]

Open solution

Q218. JEE Main 2021 (Online) 26th August Morning Shift

An aqueous KCl solution of density 1.20 g mL^{-1} has a molality of 3.30 mol kg^{-1}. The molarity of the solution in mol L^{-1} is _____ (Nearest integer) [Molar mass of KCl = 74.5]

Open solution

Q219. JEE Main 2021 (Online) 27th July Evening Shift

$2SO_2(g) + O_2(g) \rightarrow 2SO_3(g)$

The above reaction is carried out in a vessel starting with partial pressure P_{SO_2} = 250 m bar, P_{O_2} = 750 m bar and P_{SO_3} = 0 bar. When the reaction is complete, the total pressure in the reaction vessel is _____ m bar. (Round off of the nearest integer).

Open solution

Q220. JEE Main 2021 (Online) 27th July Morning Shift

The density of NaOH solution is 1.2 g cm^{-3}. The molality of this solution is _____ m. (Round off to the Nearest Integer)

[Use : Atomic mass : Na : 23.0 u, O : 16.0 u, H : 1.0 u, Density of H_2O : 1.0 g cm^{-3}]

Open solution

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Q221. JEE Main 2021 (Online) 25th July Morning Shift

When 10 mL of an aqueous solution of Fe^{2+} ions was titrated in the presence of dil H_2SO_4 using diphenylamine indicator, 15 mL of 0.02 M solution of $K_2Cr_2O_7$ was required to get the end point. The molarity of the solution containing Fe^{2+} ions is $x \times 10^{-2}$ M. The value of x is _____. (Nearest integer)

Open solution

Q222. JEE Main 2021 (Online) 25th July Morning Shift

Consider the complete combustion of butane, the amount of butane utilized to produce 72.0 g of water is _____ $\times 10^{-1}$ g. (in nearest integer)

Open solution

Q223. JEE Main 2021 (Online) 22th July Evening Shift

If the concentration of glucose ($C_6H_{12}O_6$) in blood is 0.72 g L^{-1}, the molarity of glucose in blood is _____ $\times 10^{-3}$ M. (Nearest integer)

[Given : Atomic mass of C = 12, H = 1, O = 16 u]

Open solution

Q224. JEE Main 2021 (Online) 20th July Evening Shift

4g equimolar mixture of NaOH and Na_2CO_3 contains x g of NaOH and y g of Na_2CO_3. The value of x is _____ g. (Nearest integer)

Open solution

Q225. JEE Main 2021 (Online) 20th July Morning Shift

250 mL of 0.5 M NaOH was added to 500 mL of 1 M HCl. The number of unreacted HCl molecules in the solution after complete reaction is _____ $\times 10^{21}$. (Nearest integer)

($N_A = 6.022 \times 10^{23}$)

Open solution

Q226. JEE Main 2021 (Online) 18th March Evening Shift

10.0 mL of Na_2CO_3 solution is titrated against 0.2 M HCl solution. The following titre values were obtained in 5 readings :

4.8 mL, 4.9 mL, 5.0 mL, 5.0 mL and 5.0 mL

Based on these readings, and convention of titrimetric estimation the concentration of Na_2CO_3 solution is _____ mM.

(Round off to the Nearest Integer).

Open solution

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Q227. JEE Main 2021 (Online) 18th March Morning Shift

_____ grams of 3-Hydroxy propanal (MW = 74) must be dehydrated to produce 7.8 g of acrolein (MW = 56) (C_3H_4O) if the percentage yield is 64. (Round off to the Nearest Integer).

[Given : Atomic masses : C : 12.0 u, H : 1.0 u, O : 16.0 u]

Open solution

Q228. JEE Main 2021 (Online) 18th March Morning Shift

A reaction of 0.1 mole of Benzylamine with bromomethane gave 23 g of Benzyl trimethyl ammonium bromide. The number of moles of bromomethane consumed in this reaction are $n \times 10^{-1}$, when n = _____. (Round off to the Nearest Integer).

[Given : Atomic masses : C : 12.0 u, H : 1.0 u, N : 14.0 u, Br : 80.0 u]

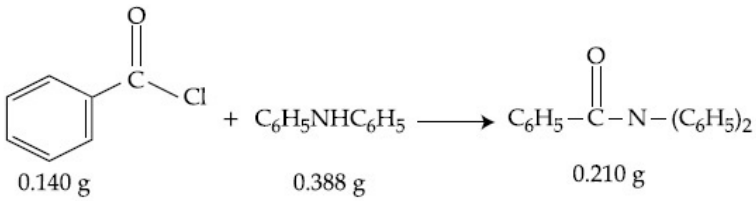
Open solution

Q229. JEE Main 2021 (Online) 18th March Morning Shift

Complete combustion of 3g of ethane gives $x \times 10^{22}$ molecules of water. The value of x is _____. (Round off to the Nearest Integer). [Use : $N_A = 6.023 \times 10^{23}$; Atomic masses in u : C : 12.0; O : 16.0; H : 1.0]

Open solution

Q230. JEE Main 2021 (Online) 17th March Evening Shift



Consider the above reaction. The percentage yield of amide product is _____. (Round off to the Nearest Integer).

(Given : Atomic mass : C : 12.0 u, H : 1.0 u, N : 14.0 u, O : 16.0 u, Cl : 35.5 u)

Open solution

Q231. JEE Main 2021 (Online) 17th March Morning Shift

15 mL of aqueous solution of Fe^{2+} in acidic medium completely reacted with 20 mL of 0.03 M aqueous $Cr_2O_7^{2-}$. The molarity of the Fe^{2+} solution is _____ $\times 10^{-2}$ M. (Round off to the Nearest Integer).

Open solution

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Q232. JEE Main 2021 (Online) 17th March Morning Shift

The mole fraction of a solute in a 100 molal aqueous solution is _____ $\times 10^{-2}$. (Round off to the Nearest Integer).

[Given : Atomic masses : H : 1.0 u, O : 16.0 u]

Open solution

Q233. JEE Main 2021 (Online) 16th March Evening Shift

When 35 mL of 0.15 M lead nitrate solution is mixed with 20 mL of 0.12 M chromic sulphate solution, _____ $\times 10^{-5}$ moles of lead sulphate precipitate out. (Round off to the Nearest Integer).

Open solution

Q234. JEE Main 2021 (Online) 16th March Morning Shift

Complete combustion of 750 g of an organic compound provides 420 g of CO_2 and 210 g of H_2O. The percentage composition of carbon and hydrogen in organic compound is 15.3 and _____ respectively. (Round off to the Nearest Integer).

Open solution

Q235. JEE Main 2021 (Online) 16th March Morning Shift

$$2MnO_4^- + bC_2O_4^{2-} + cH^+ \rightarrow xMn^{2+} + yCO_2 + zH_2O$$

If the above equation is balanced with integer coefficients, the value of c is _____. (Round off to the Nearest Integer).

Open solution

Q236. JEE Main 2021 (Online) 16th March Morning Shift

A 6.50 molal solution of KOH (aq.) has a density of 1.89 g cm^{-3}. The molarity of the solution is _____ mol dm^{-3}. (Round off to the Nearest Integer).

[Atomic masses : K : 39.0 u; O : 16.0 u; H : 1.0 u]

Open solution

Q237. JEE Main 2021 (Online) 26th February Evening Shift

The $NaNO_3$ weighed out to make 50 mL of an aqueous solution containing 70.0 mg Na^+ per mL is _____ g. (Rounded off to the nearest integer)

[Given : Atomic weight in g mol^{-1} - Na : 23; N : 14; O : 16]

Open solution

Q238. JEE Main 2021 (Online) 25th February Morning Shift

In basic medium CrO_4^{2-} oxidises $S_2O_3^{2-}$ to form SO_4^{2-} and itself changes into $Cr(OH)_4^-$. The volume of 0.154 M CrO_4^{2-} required to react with 40 mL of 0.25 M $S_2O_3^{2-}$ is _____ mL. (Rounded off to the nearest integer)

Open solution

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Q239. JEE Main 2021 (Online) 24th February Evening Shift

The formula of a gaseous hydrocarbon which requires 6 times of its own volume of O_2 for complete oxidation and produces 4 times its own volume of CO_2 is C_xH_y. The value of y is _____.

Open solution

Q240. JEE Main 2021 (Online) 24th February Evening Shift

1.86 g of aniline completely reacts to form acetanilide. 10% of the product is lost during purification. Amount of acetanilide obtained after purification (in g) is _____ $\times 10^{-2}$.

Open solution

Q241. JEE Main 2021 (Online) 24th February Morning Shift

4.5 g of compound A (MW = 90) was used to make 250 mL of its aqueous solution. The molarity of the solution in M is x $\times 10^{-1}$.
The value of x is _____. (Rounded off to the nearest integer)

Open solution

Q242. JEE Main 2020 (Online) 4th September Evening Slot

Consider the following equations :
$2Fe^{2+} + H_2O_2 \rightarrow xA + yB$
(in basic medium)
$2MnO_4^- + 6H^+ + 5H_2O_2 \rightarrow x'C + y'D + z'E$
(in acidic medium)
The sum of the stoichiometric coefficients x, y, x', y', and z' for products A, B, C, D and E, respectively, is _____.

Open solution

Q243. JEE Main 2020 (Online) 4th September Morning Slot

The mass of ammonia in grams produced when 2.8 kg of dinitrogen quantitatively reacts with 1 kg of dihydrogen is _____.

Open solution

Q244. JEE Main 2020 (Online) 4th September Morning Slot

A 20.0 mL solution containing 0.2 g impure H_2O_2 reacts completely with 0.316 g of $KMnO_4$ in acid solution. The purity of H_2O_2 (in %) is _____
(mol. wt. of H_2O_2 = 34; mol. wt. of $KMnO_4$ = 158)

Open solution

Q245. JEE Main 2020 (Online) 3rd September Evening Slot

The volume (in mL) of 0.1 N NaOH required to neutralise 10 mL of 0.1 N phosphinic acid is _____.

Open solution

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Q246. JEE Main 2020 (Online) 3rd September Evening Slot

6.023×10^{22} molecules are present in 10 g of a substance 'x'. The molarity of a solution containing 5 g of substance 'x' in 2 L solution is _____ $\times 10^{-3}$

Open solution

Q247. JEE Main 2020 (Online) 3rd September Morning Slot

The mole fraction of glucose ($C_6H_{12}O_6$) in an aqueous binary solution is 0.1. The mass percentage of water in it, to the nearest integer, is _____.

Open solution

Q248. JEE Main 2020 (Online) 3rd September Morning Slot

The volume strength of 8.9 M H_2O_2 solution calculated at 273 K and 1 atm is _____. (R = 0.0821 L atm K^{-1} mol^{-1}) (rounded off ot the nearest integer)

Open solution

Q249. JEE Main 2020 (Online) 2nd September Evening Slot

The ratio of the mass percentages of 'C & H' and 'C & O' of a saturated acyclic organic compound 'X' are 4 : 1 and 3 : 4 respectively. Then, the moles of oxygen gas required for complete combustion of two moles of organic compound 'X' is _____.

Open solution

Q250. JEE Main 2020 (Online) 9th January Evening Slot

10.30 mg of O_2 is dissolved into a liter of sea water of density 1.03 g/mL. The concentration of O_2 in ppm is_____.

Open solution

Q251. JEE Main 2020 (Online) 9th January Morning Slot

The hardness of a water sample containing 10^{-3} M $MgSO_4$ expressed as $CaCO_3$ equivalents (in ppm) is _____.
(molar mass of $MgSO_4$ is 120.37 g/mol)

Open solution

Q252. JEE Main 2020 (Online) 9th January Morning Slot

The molarity of HNO_3 in a sample which has density 1.4 g/mL and mass percentage of 63% is _____.
(Molecular Weight of HNO_3 = 63)

Open solution

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Q253. JEE Main 2020 (Online) 8th January Morning Slot

The volume (in mL) of 0.125 M $AgNO_3$ required to quantitatively precipitate chloride ions in 0.3 g of $[Co(NH_3)_6]Cl_3$ is _____.

$M_{[Co(NH_3)_6Cl_3]}$ = 267.46 g/mol

M_{AgNO_3} = 169.87 g/mol

Open solution

Q254. JEE Main 2020 (Online) 8th January Morning Slot

Ferrous sulphate heptahydrate is used to fortify foods with iron. The amount (in grams) of the salt required to achieve 10 ppm of iron in 100 kg of wheat is _____.

Atomic weight : Fe = 55.85; S = 32.00; O = 16.00

Open solution

Out of syllabus 5 questions

These questions are on topics no longer in the current syllabus. They are kept for practice, after everything else.

Q255. JEE Main 2023 (Online) 8th April Evening Shift · MCQ (Single Correct Answer)

Which of the following have same number of significant figures?

A. 0.00253

B. 1.0003

C. 15.0

D. 163

Choose the correct answer from the options given below

(A) A, C and D only (B) B and C only (C) A, B and C only (D) C and D only

Open solution

Q256. JEE Main 2022 (Online) 29th June Evening Shift · MCQ (Single Correct Answer)

Using the rules for significant figures, the correct answer for the expression $\frac{0.02858\times0.112}{0.5702}$ will be

(A) 0.005613 (B) 0.00561 (C) 0.0056 (D) 0.006

Open solution

Q257. JEE Main 2021 (Online) 25th July Evening Shift · Numerical

The number of significant figures in 0.00340 is _____.

Open solution

Q258. JEE Main 2021 (Online) 26th February Morning Shift · Numerical

The number of significant figures in 50000.020×10^{-3} is _____.

Open solution

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Q259. JEE Main 2020 (Online) 8th January Evening Slot · **Numerical**

Open solution

$NaClO_3$ is used, even in spacecrafts, to produce O_2. The daily consumption of pure O_2 by a person is 492L at 1 atm, 300K. How much amount of $NaClO_3$, in grams, is required to produce O_2 for the daily consumption of a person at 1 atm, 300 K?

$NaClO_3(s) + Fe(s) \rightarrow O_2(g) + NaCl(s) + FeO(s)$

$R = 0.082\ L\ atm\ mol^{-1}\ K^{-1}$

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Answer Key

MCQ (Single Correct Answer)

1. **C**	2. **D**	3. **B**	4. **C**	5. **B**
6. **D**	7. **D**	8. **B**	9. **B**	10. **D**
11. **A**	12. **D**	13. **D**	14. **B**	15. **C**
16. **B**	17. **A**	18. **B**	19. **D**	20. **B**
21. **A**	22. **C**	23. **C**	24. **C**	25. **A**
26. **D**	27. **B**	28. **A**	29. **C**	30. **D**
31. **D**	32. **B**	33. **B**	34. **A**	35. **B**
36. **D**	37. **B**	38. **C**	39. **B**	40. **A**
41. **B**	42. **C**	43. **B**	44. **D**	45. **C**
46. **A**	47. **C**	48. **C**	49. **A**	50. **C**
51. **D**	52. **C**	53. **B**	54. **C**	55. **C**
56. **C**	57. **D**	58. **C**	59. **D**	60. **C**
61. **B**	62. **D**	63. **C**	64. **B**	65. **D**
66. **B**	67. **C**	68. **D**	69. **C**	70. **C**
71. **A**	72. **B**	73. **A**	74. **A**	75. **B**
76. **C**	77. **C**	78. **C**	79. **B**	80. **C**
81. **C**	82. **B**	83. **A**	84. **A**	85. **D**
86. **C**	87. **C**	88. **A**	89. **B**	90. **A**
91. **C**	92. **B**	93. **A**	94. **C**	95. **B**
96. **B**	97. **B**	98. **D**	99. **B**	100. **B**
101. **D**	102. **B**	103. **A**	104. **A**	105. **C**
106. **D**	107. **B**	108. **D**	109. **C**	110. **A**
111. **D**	112. **C**	113. **A**	114. **A**	115. **D**
116. **D**	117. **B**	118. **D**	119. **A**	120. **D**
121. **A**	122. **B**	123. **B**	124. **B**	125. **A**
126. **A**	127. **D**	128. **C**	129. **C**	130. **A**

Numerical

131. **73**	132. **1825**	133. **1**	134. **138**	135. **12**
136. **65**	137. **2**	138. **63**	139. **9**	140. **3**
141. **80**	142. **125**	143. **1655**	144. **420**	145. **61**

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146. **100**	147. **153**	148. **57**	149. **164**	150. **22**
151. **74**	152. **95**	153. **45**	154. **14**	155. **24**
156. **11**	157. **50**	158. **78**	159. **7**	160. **11**
161. **4**	162. **815**	163. **4**	164. **135**	165. **7**
166. **8**	167. **4**	168. **100**	169. **56**	170. **1**
171. **224**	172. **36**	173. **11**	174. **1**	175. **139**
176. **364**	177. **227**	178. **59**	179. **4**	180. **44**
181. **150**	182. **148**	183. **15**	184. **200**	185. **12**
186. **12**	187. **5**	188. **4**	189. **180**	190. **4**
191. **6**	192. **25**	193. **2**	194. **46**	195. **24**
196. **3**	197. **24**	198. **54**	199. **46**	200. **33**
201. **3810**	202. **46**	203. **2**	204. **2**	205. **143**
206. **225**	207. **18**	208. **25**	209. **5418**	210. **0**
211. **1143**	212. **2**	213. **64**	214. **13**	215. **20**
216. **19**	217. **50**	218. **3**	219. **875**	220. **5**
221. **18**	222. **464**	223. **4**	224. **1**	225. **226**
226. **50**	227. **16**	228. **3**	229. **18**	230. **77**
231. **24**	232. **64**	233. **525**	234. **3**	235. **16**
236. **9**	237. **13**	238. **173**	239. **8**	240. **243**
241. **2**	242. **19**	243. **3400**	244. **85**	245. **10**
246. **25**	247. **47**	248. **100**	249. **5**	250. **10**
251. **100**	252. **14**	253. **26.80 to 27.00**	254. **4.95 to 4.97**	

Out of syllabus

255. **A**	256. **B**	257. **3**	258. **8**	259. **2120 to 2140**

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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